A project report on

EEG-BASED EXOSKELETON CONTROL FOR UPPER- AND LOWER-LIMB NEUROREHABILITATION

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Robotics)

by

PARDEEP SINGH (23BRS1123)

AASTHA KUMARI (23BRS1040)



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Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

November, 2026

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DECLARATION

I hereby declare that the thesis entitled “EEG-BASED EXOSKELETON CONTROL FOR UPPER- AND LOWER-LIMB NEUROREHABILITATION” submitted by me, PARDEEP SINGH (Register No. 23BRS1123), for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Robotics), Vellore Institute of Technology, Chennai is a record of bonafide work carried out by me under the supervision of my Project Guide, School of Computer Science and Engineering, Vellore Institute of Technology, Chennai.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Chennai

Date: 13th October 2026

Signature of the Candidate:

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School of Computer Science and Engineering

CERTIFICATE

This is to certify that the report entitled “EEG-BASED EXOSKELETON CONTROL FOR UPPER- AND LOWER-LIMB NEUROREHABILITATION” is prepared and submitted by PARDEEP SINGH (Register No: 23BRS1123) to Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Robotics) is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

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Date:

ABSTRACT

Post-stroke motor impairments and spinal cord injuries afflict over 15 million individuals annually, necessitating prolonged neurorehabilitation to stimulate neuroplastic motor recovery. While brain-computer interfaces (BCIs) paired with robotic exoskeletons have shown immense clinical efficacy, conventional solutions remain confined to specialized laboratories due to cumbersome 64-channel wet-electrode scalp caps, complex multi-minute calibration protocols, and severe vulnerability to ocular and motion artifacts. This project presents the design, mathematical formulation, edge-AI deployment, and verification of a unified embedded BCI architecture for cross-limb neurorehabilitation that entirely eliminates scalp caps in favor of a discreet, around-the-ear electrode array (OpenBCI Open-cEEGrid, 16 channels, 250 Hz).

The architecture implements a four-tier processing hierarchy connecting biosignal acquisition to real-time exoskeleton actuation. At the edge host level (NVIDIA Jetson Orin Nano), incoming telemetry is conditioned through a zero-phase 4th-order Butterworth bandpass filter (0.5–45.0 Hz) and a 50 Hz IIR notch filter (Q=30) operating over a thread-safe circular ring buffer (2.0s window, 100 ms step). Feature extraction concurrently decodes four distinct biological intent signals: (1) time-locked tactile/auditory P300 Event-Related Potentials (ERP) within the 250–500 ms post-stimulus window; (2) relative log-power alpha rhythms (8–13 Hz) for cognitive workload monitoring; (3) temporalis muscle high-gamma bursts (30–45 Hz) for deliberate jaw-clench discrete triggering; and (4) periauricular differential electrooculography (EOG) for immediate double-blink safety veto.

Classification is executed using Stepwise Linear Discriminant Analysis (SWLDA) with statistical forward feature selection, alongside LDA for workload discrimination. A stateful priority arbitrator resolves conflicting intents via a strict five-level hierarchy, enforcing a 300 ms dwell-time confirmation gate to suppress involuntary triggers and an immediate 500 ms lockout upon blink-veto detection. Validated intents are serialized into a robust 7-byte binary UART frame protected by CRC-8/MAXIM checksums and transmitted at 115200 baud to an ESP32-P4 microcontroller. Dual-core FreeRTOS firmware coordinates trapezoidal trajectory

generation across PCA9685 PWM servo drivers (5-DOF cable-driven hand module) and dual L298N H-bridge motor controllers (2-DOF discrete-stepping leg module).

Rigorous experimental validation across benchmark physiological datasets and synthetic streams demonstrates 86.0% classification accuracy (AUC = 0.884) for P300 intent detection, 96.5% accuracy for jaw-clench triggering, and a 12.5 ms response time for double-blink emergency veto. In-vivo validation on the wireless 8-channel Qneuro H100 headset with 57,250 human electrophysiological samples achieved 87.0% accuracy in 3-class motor intent classification (Rest vs. Hand vs. Leg), 83.8% in 5-class lateralized classification, and 100% sensitivity with 0% false alarms for emergency double-blink veto and jaw-clench safety interrupts. The total computational pipeline consumes only 9.81 ms per cycle, operating with a 10.2× headroom factor within the 100 ms soft real-time budget. The resulting testbed provides a viable, user-friendly, and portable cross-limb neurorehabilitation platform.

Keywords: Brain-Computer Interface (BCI), Around-the-Ear EEG, cEEGrid, Exoskeleton Control, Neurorehabilitation, Stepwise LDA, Edge-AI, Jetson Orin Nano, ESP32-P4, Emergency Blink-Veto, Qneuro H100, Motor Intent Classification.

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Place: **Chennai**

Date: **13th** **October** **2026**

PARDEEP SINGH (23BRS1123)

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Ground Truth Cues,
Interrupts, and Commands

LIST OF SYMBOLS AND ABBREVIATIONS

BCI	Brain-Computer Interface
cEEGrid	Continuous Around-the-Ear Circular Grid (16-channel flex-printed sensor)
EEG	Electroencephalography
EOG	Electrooculography (corneal-retinal ocular potential recording)
EMG	Electromyography (muscle activation bioelectric potential)
ERD	Event-Related Desynchronization (power suppression during motor engagement)
ERP	Event-Related Potential (time-locked neuroelectric response)
ERS	Event-Related Synchronization (post-movement power rebound)
SWLDA	Stepwise Linear Discriminant Analysis
LDA	Linear Discriminant Analysis
RF	Random Forest (ensemble bagging classifier)
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
PR-AUC	Area Under the Precision-Recall Curve
PSD	Power Spectral Density (Welch periodogram, dB/Hz)
STFT	Short-Time Fourier Transform
IIR	Infinite Impulse Response (digital filter topology)
FIR	Finite Impulse Response
CRC	Cyclic Redundancy Check (CRC-8/MAXIM, polynomial 0x31)
UART	Universal Asynchronous Receiver-Transmitter (115200 baud serial)
LSL	Lab Streaming Layer (time-synchronized biosignal networking protocol)
SPP	Serial Port Profile (Bluetooth RFCOMM transport)
PPG	Photoplethysmography (optical pulse/heart-rate sensor)
MCU	Microcontroller Unit (ESP32-P4 Dual-Core RISC-V)
FreeRTOS	Real-Time Operating System for Embedded

	Microcontrollers
PWM	Pulse Width Modulation (servo motor angle driver)
DOF	Degrees of Freedom (mechanical articulation joints)
FSR	Force-Sensitive Resistor (tactile fingertip sensor)
SCI	Spinal Cord Injury
FMA	Fugl-Meyer Assessment (standardized motor recovery scoring instrument)
CO	Course Outcome (ABET / NBA academic accreditation rubric)

CHAPTER 1
INTRODUCTION

1.1 CLINICAL BACKGROUND AND MOTIVATION

Stroke and traumatic neurotrauma represent the leading causes of chronic adult physical disability worldwide. According to epidemiological surveillance by the World Health Organization and the American Heart Association (Park et al., 2025), approximately 15 million individuals suffer an acute stroke each year. Of these, over 5 million survive with permanent unilateral motor deficits, most commonly hemiparesis characterized by flaccid or spastic loss of voluntary control over upper-extremity distal grasping and lower-extremity gait locomotion.

Contemporary neurorehabilitation science has established that the human central nervous system retains remarkable adaptive plasticity following focal cerebral ischemia. However, promoting functional cortical reorganization requires intensive, repetitive, task-specific, and patient-engaged physical training (Sieghartsleitner et al., 2024). Traditional manual physiotherapy is physically demanding for clinicians, expensive for healthcare systems, and inherently unscalable. Robotic exoskeletons offer an automated medium for high-dosage physical therapy, but passive robotic motion that drives a limb without patient neural involvement yields significantly inferior motor relearning compared to active, intention-driven therapy (Park et al., 2025).

Brain-Computer Interfaces (BCIs) address this critical bottleneck by directly decoding central nervous system motor intentions and closing the sensorimotor loop through real-time robotic actuation. When a patient attempts or imagines movement, the decoded neuroelectric potential immediately triggers the exoskeleton to execute the physical movement, reinforcing hebbian neuroplasticity: ‘neurons that fire together, wire together’ (Birbaumer et al., 2008).

1.2 PROBLEM STATEMENT

Despite compelling clinical trials demonstrating that BCI-driven robotic rehabilitation significantly accelerates motor recovery compared to conventional therapy alone, current BCI systems remain largely confined to specialized research institutes and

tertiary hospitals. The fundamental limitations preventing widespread bedside and home-based adoption include:

1. Cumbersome Scalp Caps: Conventional BCIs rely on 32- to 64-channel wet-electrode caps distributed across the international 10–20 system. Applying conductive gel requires 30 to 45 minutes of preparation by trained personnel, causes scalp irritation, dries out during extended sessions, and is socially stigmatizing for patients.

2. Long Calibration and Decoupled Modalities: Existing systems require extensive subject-specific calibration before each session. Furthermore, commercial systems are strictly siloed—operating either exclusively for upper-limb hand rehabilitation or exclusively for lower-limb gait retraining, requiring clinics to purchase and maintain separate, incompatible equipment.

3. Vulnerability to Motion and Ocular Artifacts: High-impedance scalp caps pick up severe electromyographic (EMG) and electrooculographic (EOG) interference during patient movement, frequently triggering unintended robotic movements.

4. Absence of Deterministic Safety Interlocks: Standard laboratory BCIs prioritize classification accuracy over fail-safe real-time shutdown. In physical human-robot interaction, an unintended actuation can cause musculoskeletal strain or patient injury.

1.3 LIMITATIONS OF CONVENTIONAL SCALP EEG

Traditional EEG-based BCIs encounter several critical bottlenecks that prevent practical clinical and domestic translation:

1. Bulky Whole-Scalp Caps: Standard 64-channel 10–20 caps require conductive electrolytic gel application, causing hair contamination, long setup times (>30–45 minutes), and impedance degradation within 1–2 hours.

2. Motor Imagery (MI) Inefficiency: Classical Sensorimotor Rhythm (SMR) / Motor Imagery BCIs require extensive user training (often dozens of sessions), and suffer from “BCI illiteracy” in 15–30% of stroke survivors whose motor cortices are severely infarcted.

3. Rigid Edge Processing Constraints: Clinical rehabilitation exoskeletons require sub-100 ms real-time responsiveness and rigorous fail-safe emergency veto mechanisms. Desktop-tethered pipelines cannot operate in ambulatory, mobile rehabilitation environments.

1.4 THE AROUND-THE-EAR (CEEGRID) PARADIGM

To overcome the mechanical and clinical barriers of full-scalp caps, this project utilizes the cEEGrid periauricular electrode array (Knierim et al., 2021, 2022). Developed as a flexible printed circuit board strip that conforms naturally around the ear, the cEEGrid provides 16 localized electrodes (8 per ear: L1–L8 and R1–R8) positioned over the temporal and periauricular cranial surface. Operating with an OpenBCI Cyton+Daisy biosensing amplifier at 250 Hz, the array offers distinct clinical advantages:

• Rapid Application: The adhesive flex-strip can be applied in under 3 minutes using minimal electrolyte paste, requiring zero assistance from specialized EEG technicians.

• Patient Comfort & Concealment: The periauricular footprint can be integrated into ordinary headphone form-factors (King et al., 2025), preserving patient dignity and eliminating social stigma.

• Multi-Biosignal Surface: Beyond cerebral electroencephalography, the temporal electrode placement provides pristine proximity to ocular dipoles for periauricular EOG tracking and temporalis muscle recruitment for high-gamma EMG detection.

1.5 DUAL-MODULE CROSS-LIMB CONTROL PHILOSOPHY

While the cEEGrid eliminates scalp caps, its anatomical placement around the temporal lobe means it lacks coverage over the central primary motor cortex (C3, Cz, C4). Consequently, traditional continuous Motor Imagery (MI) sensorimotor rhythm decoding is physiologically unfeasible (Knierim et al., 2021). To circumvent this constraint while establishing a clinically effective cross-limb rehabilitation platform, this project adopts a discrete event-triggered control paradigm:

• Upper-Limb Hand Module: Grasp opening is triggered via oddball P300 Event-Related Potentials (ERPs) validated on periauricular channels (Eidel et al., 2024), while grasp closing is commanded via deliberate isometric temporalis muscle jaw-clenching.

• Lower-Limb Leg Module: Gait stepping states (step forward / step back) are sequenced via confirmed discrete neuroelectric commands, supported by cognitive workload gating.

• Multi-Tier Safety Layer: An immediate, deterministic emergency stop (L1 Veto) is mapped to periauricular double-blink EOG signals, providing sub-15ms physical motor shutdown.

1.6 RESEARCH OBJECTIVES

The core objectives of this project are:

1. Develop an open-source, thread-safe real-time signal acquisition and conditioning pipeline on an edge computing platform (NVIDIA Jetson Orin Nano) operating at 250 Hz over 16 periauricular channels.

2. Formulate and implement lightweight feature extraction algorithms capturing P300 ERP temporal windows, Welch relative bandpower, temporalis high-gamma EMG, and differential EOG dipoles.

3. Deploy Stepwise Linear Discriminant Analysis (SWLDA) with statistical forward feature selection and export sparse weight representations for sub-100µs edge execution.

4. Design a robust 7-byte binary UART communication protocol protected by CRC-8/MAXIM checksums running at 115200 baud.

5. Construct dual-core FreeRTOS firmware on an ESP32-P4 microcontroller driving physical hand and leg robotic prototypes with trapezoidal velocity profiles and safety interlocks.

6. Experimentally validate that end-to-end processing latency remains under 100 ms and resting false positive trigger rate remains below 3.0%.

7. Conduct in-vivo empirical validation using the wireless Qneuro H100 headset with real human EEG data, demonstrating movement intent classification accuracy exceeding 80% on at least 3 motor classes.

1.7 SCOPE AND TECHNICAL BOUNDARIES

This project focuses on the computational, electrical, and mechatronic validation of a functional prototype benchtop testbed using synthetic anthropomorphic hand and leg linkages, commercial OpenBCI hardware, the wireless Qneuro H100 NeuroState headset, and standardized public EEG benchmark datasets (PhysioNet eegmmidb, oddball ERP records). It does not encompass human clinical trials with acute stroke

patients, which will require formal institutional ethics review board approval and medical-grade electrical isolation certification in future work.

1.8 REPORT ORGANIZATION

The remainder of this report is structured as follows: Chapter 2 reviews the state-of-the-art across 17 recent literature publications, highlighting research gaps in cross-limb control. Chapter 3 details the four-tier system architecture, mathematical derivations, power budget, risk register, and firmware design. Chapter 4 provides extensive experimental results on benchmark datasets, analyzing 8 publication figures, latency waterfalls, and classification benchmarks. Chapter 5 presents the comprehensive in-vivo validation campaign conducted on the wireless Qneuro H100 headset with 57,250 real human electrophysiological samples, including 4 live data acquisition oscilloscope recordings and 12 analytical figures (16 figures total). Chapter 6 concludes the report and presents future clinical directions. Appendices document individual member contributions, guide feedback incorporation, source code implementations, and the complete hardware bill of materials.

1.9 CHAPTER SUMMARY

This introductory chapter established the clinical imperative for intention-driven robotic neurorehabilitation over passive therapy, framed the acute barriers of conventional 64-channel whole-scalp caps, and introduced the around-the-ear (cEEGrid) sensing paradigm. A discrete event-triggered cross-limb control philosophy was formulated, uniting an oddball P300 and jaw-clench hand orthosis with a discrete-stepping leg linkage. Seven concrete technical objectives and scope boundaries were delineated to guide the engineering design and validation presented in subsequent chapters.

CHAPTER 2
LITERATURE REVIEW

2.1 THEORETICAL FOUNDATIONS OF BCI NEUROREHABILITATION

The therapeutic efficacy of Brain-Computer Interfaces in neurorehabilitation rests upon the closed-loop coupling of cortical motor intent with peripheral sensory feedback (Birbaumer et al., 2008). In a landmark umbrella review of systematic reviews examining robot-assisted upper limb stroke rehabilitation, Park et al. (2025) synthesized evidence across 48 clinical trials, concluding that robotic therapy incorporating active patient intent significantly improves Fugl-Meyer Assessment (FMA) motor scores compared to passive robotic motion or conventional physical therapy alone. Similarly, Sieghartsleitner et al. (2024) demonstrated in the recoveriX clinical protocol that sequential training of the upper extremity followed by the lower extremity yields cross-limb neuroplastic transfer, improving gait velocity and hand dexterity concurrently.

The neurophysiological basis for closed-loop BCI rehabilitation lies in Hebbian synaptic plasticity and long-term potentiation (LTP). When volitional cortical motor signals temporally coincide with kinesthetic/proprioceptive sensory feedback from robotic limb movement, the redundant sensorimotor neural circuits undergo structural strengthening. Wolpaw et al. (2002) established the canonical BCI framework outlining signal acquisition, feature extraction, translation, and device output as an operant conditioning process driving cortical adaptation.

2.2 TAXONOMY OF BIOSIGNAL MODALITIES: MI VS ERP VS PERIAURICULAR SIGNALS

Motor Imagery (MI) based on sensorimotor rhythm Event-Related Desynchronization/Synchronization (ERD/ERS) has traditionally served as the primary paradigm for neurorehabilitation BCIs (Zare et al., 2025; Zhao et al., 2024). However, as demonstrated by De la Cruz-Sánchez et al. (2025), robust MI classification requires electrodes placed directly over the motor cortex (C3, Cz, C4). When electrodes are shifted to the periauricular temporal region, sensorimotor ERD attenuates below detectable signal-to-noise thresholds (Knierim et al., 2021).

Conversely, Event-Related Potentials—specifically the P300 oddball response—are widely distributed across temporal and parietal cortices. Eidel et al. (2024) successfully demonstrated that tactile and auditory P300 potentials can be recorded reliably via the around-the-ear cEEGrid array, achieving single-trial classification accuracies of 63–68%, significantly above chance level. The foundational taxonomy of the hybrid BCI formulated by Pfurtscheller et al. (2006) established that fusing distinct biosignal streams markedly improves decoding reliability while reducing unintended triggers. Building on this framework, Zhu et al. (2020) and King et al. (2025) validated that hybrid systems combining electroencephalography, electrooculography (EOG), and electromyography (EMG) achieve dramatically superior reliability and safety compared to unimodal EEG systems.

Recent advances in deep learning for EEG classification, particularly the Convolutional Transformer Network (CTNet) proposed by Zhao et al. (2024), have demonstrated impressive motor imagery decoding accuracies exceeding 91% on high-density 64-channel scalp recordings. However, such architectures require substantial GPU compute resources and multi-million parameter models, rendering them unsuitable for real-time edge deployment on low-power embedded platforms without significant model compression. The cognitive load detection framework proposed by Yedukondalu et al. (2024) using time-frequency EEG features with Bi-LSTM networks provides complementary insights into fatigue monitoring, though its computational overhead similarly exceeds embedded budgets.

2.3 AROUND-THE-EAR EEG (CEEGRID) STATE-OF-THE-ART

The cEEGrid sensor array, pioneered by Debener and Bleichner, was thoroughly characterized for open-source biosensing by Knierim et al. (2021, 2022). Their benchtop characterization verified that the 16-electrode periauricular array maintains skin-electrode contact impedance below 15 kΩ when prepared with minimal electrolyte gel, sustaining stable signal recording over 6-hour continuous trials. In a breakthrough 2025 Nature Scientific Reports publication, King et al. (2025) proved that periauricular electrodes can track horizontal and vertical ocular movements (earEOG) with sub-degree accuracy by measuring differential corneal-retinal dipole rotations, directly enabling discrete emergency-stop veto gating in mobile headwear.

Periauricular arrays position electrodes around the pinna (auricle), capturing: (1) Temporal and parietal cortical activity: adequate projection of P300 ERPs and alpha-band oscillations; (2) Periauricular differential EOG: dipolar ocular potentials during vertical and horizontal saccades and blinks; and (3) Temporalis EMG bursts: high-frequency electromyographic signatures (>30 Hz) during intentional mandible clenches. This multi-biosignal sensitivity from a single compact electrode strip is the key advantage exploited in this project.

2.4 UPPER- AND LOWER-LIMB EXOSKELETON ARCHITECTURES

Robotic orthoses for motor restoration are broadly divided into cable-driven soft gloves and rigid articulated linkages. Araujo et al. (2021) introduced HERO, a low-cost 3D-printed hand exoskeleton utilizing flexible Bowden cable tendons driven by hobby servomotors, demonstrating effective finger flexion-extension. Triwiyanto et al. (2023, 2025) developed bilateral 3D-printed upper-limb exoskeletons integrated with IMUs and force-sensitive resistors (FSR), establishing that closed-loop force feedback is vital to prevent hyperextension injuries. For the lower limb, Ma et al. (2024) and Wang et al. (2023) developed gait-assistive exoskeletons utilizing digital twin state machines, demonstrating that discrete event triggering (start gait, stop gait, sit-to-stand) provides far higher patient stability than continuous velocity control.

In a comprehensive survey on exoskeleton safety hazards across 84 clinical centers, Massardi et al. (2023) identified that unintended actuator triggers and failure to stop during patient spasms constitute the highest-risk failure modes. The study emphasized that pure software BCIs without redundant hardware interlocks are clinically unacceptable, underscoring the necessity of a layered safety architecture where physiological veto mechanisms override autonomous controller states.

2.5 EMBEDDED CLASSIFIERS AND EDGE INTELLIGENCE

Deploying BCI algorithms on mobile edge hardware introduces severe compute and thermal constraints. While deep learning architectures like CTNet (Zhao et al., 2024) achieve exceptional motor imagery decoding on high-end desktop GPUs, their multi-million parameter footprints require substantial memory bandwidth and induce inference latencies exceeding 200 ms on embedded systems. In contrast, Stepwise Linear Discriminant Analysis (SWLDA) performs statistical feature down-selection,

yielding sparse linear projection vectors that can be evaluated via simple dot-products ($w \cdot x + b$) on low-power microcontrollers in under 50 microseconds without quantization degradation (Yedukondalu et al., 2024).

Modern rehabilitation robotics mandates real-time safety compliance (IEC 60601-1 / ISO 13482). Classical cloud or PC-tethered pipelines exhibit high jitter (30–120 ms) and communication drops. Distributed architectures segregating heavy neural feature pipelines on embedded edge processors (NVIDIA Jetson) and deterministic real-time PWM motion profiling on dual-core microcontrollers (ESP32 FreeRTOS) represent the state of the art in medical cyber-physical systems.

2.6 MULTI-TIER SAFETY INTERLOCKS AND CLINICAL RISK MITIGATION

Clinical translation of active neuro-assistive orthoses is strictly governed by medical safety standards, principally ISO 14971 (Application of risk management to medical devices), ISO 13482 (Safety requirements for personal care and physical assistant robots), and IEC 60601-1 (Medical electrical equipment safety). To satisfy ISO 13482 constraints, safety architecture cannot rely on software watchdog routines alone; it mandates deterministic multi-tiered hardware interlocks, immediate biological veto channels (< 15 ms latency), mechanical hard stops, and hardwired power-relay cutoffs. Coupling biological veto gating (periauricular double-blink earEOG) directly to MCU interrupt lines provides fail-safe clinical containment against involuntary spasms or neural decoder glitches.

2.7 LITERATURE SYNTHESIS MATRIX (17 CORE PUBLICATIONS)

Table 2.1 summarizes the 17 core publications reviewed, detailing their biosignal modality, target hardware, classification approach, and limitations.

Table 2.1: Comprehensive Literature Synthesis Matrix (17 Core Publications)

S. No.	Author & Year	Biosignal / Paradigm	Hardware Platform	Classifier	Key Findings & Limitations
1	Zare et al. (2025)	EEG Motor Imagery	Soft Robotic Glove	CNN / SVM	84.2% MI accuracy; relies on 32-ch wet scalp cap.
2	Araujo et al. (2021)	EEG Motor Imagery	HERO Hand Exoskeleton	LDA	Low-cost textile 3D print; long setup time (>30 min).
3	Triwiyanto (2023)	IMU Wireless	Hand Exoskeleton	Rule-based	Bilateral sync; lacks neural intention decoding.
4	Triwiyanto (2025)	FSR Force Sensor	Upper Limb Exoskeleton	Thresholding	Effective grip control; purely mechanical trigger.
5	Knierim et al. (2021)	16-ch cEEGrid EEG	OpenBCI Cyton+Daisy	Benchmark PSD	Validated around-the-ear feasibility; no exoskeleton.
6	Knierim et al. (2022)	cEEGrid Adapter	HardwareX OpenBCI	Hardware Spec	Open adapter design; lacks classifier integration.
7	Eidel et al. (2024)	Tactile P300 ERP	cEEGrid Temporal	SWLDA	63-68% P300 accuracy around ear; proved ERP viability.
8	Sieghartsleitner (2024)	Upper & Lower BCI	recoveriX Commercial	CSP + LDA	Proved sequential cross-limb recovery; $40k+ cost.

S. No.	Author & Year	Biosignal / Paradigm	Hardware Platform	Classifier	Key Findings & Limitations
9	Zhao et al. (2024)	EEG Motor Imagery	CTNet Transformer	Conv-Transformer	91.2% accuracy; too heavy for low-power edge MCU.
10	Zhu et al. (2020)	Hybrid SSVEP + EOG	Robotic Manipulator	CCA + Threshold	Proved multimodal reliability; visual flicker fatigue.
11	King et al. (2025)	earEOG Periauricular	Headphone Prototype	Dipole Tracking	Validated ocular veto in ear footprint; Nature 2025.
12	Massardi et al. (2023)	Hazard Analysis	Exoskeleton Survey	Risk Assessment	Emphasized critical need for redundant hard E-stops.
13	Park et al. (2025)	Systematic Review	Upper Limb Robots	Meta-Analysis	Proved active neural intent essential for FMA recovery.
14	Yedukondalu (2024)	Cognitive Load EEG	Bi-LSTM / LDA	Time-Frequency	Alpha power tracks mental fatigue; high compute.
15	De la Cruz (2025)	EEG Spatiotemporal	Hand Exoskeleton	CSP + SVM	Validated MI trajectory; requires central C3/C4 leads.
16	Ma et al. (2024)	Lower Limb BCI	Gait Exoskeleton	Attention MI	Discrete state transitions essential for leg stability.
17	Wang et al. (2023)	Digital Twin Exoskeleton	Lower Limb Robotics	State Machine	Digital twin monitoring; no lightweight around-ear EEG.

2.8 IDENTIFICATION OF RESEARCH GAPS

From the synthesis of the 17 core publications, three glaring research gaps emerge:

1. The Scalp-Cap vs. Wearability Gap: High-accuracy BCIs in literature (Zare et al., 2025; Zhao et al., 2024) universally rely on 32- to 64-channel wet scalp caps, rendering them clinically unviable for home or bedside use. Conversely, around-the-ear EEG studies (Knierim et al., 2021; Eidel et al., 2024) have operated as pure observational neuroscience experiments without interfacing to robotic actuators.

2. The Cross-Limb Silo Gap: All surveyed systems are strictly segregated into upper-limb hand orthoses or lower-limb gait exoskeletons. Despite clinical proof that cross-limb training promotes neuroplastic recovery (Sieghartsleitner et al., 2024), no unified embedded BCI platform exists capable of dynamically switching telemetry between hand and leg testbeds on a single architecture.

3. The Safety Latency Gap: Existing BCI robotics rely on software timeouts or external push-buttons. No prior system integrates sub-15ms biological emergency veto gating directly derived from the same periauricular headset.

Table 2.2: Identification of Critical Research Gaps and Project Resolution Strategy

Research Gap	Literature Evidence	How This Project Addresses the Gap
Scalp-Cap vs. Wearability	Knierim (2021), Eidel (2024): cEEGrid observational only. Zare (2025), Zhao (2024): 32-64ch wet scalp required.	Deploys cEEGrid + Qneuro H100 periauricular headsets directly controlling robotic exoskeleton actuators.
Cross-Limb Silo	Sieghartsleitner (2024): Proved cross-limb neuroplastic transfer, but on $40k+ commercial recoveriX system.	Unified MODULE_HAND (0x01) + MODULE_LEG (0x02) architecture on single ESP32-P4 MCU.
Safety Latency	Massardi (2023): Surveyed 84 centers; unintended triggers are highest-risk failure.	Sub-15 ms double-blink EOG veto with 500 ms lockout directly from periauricular electrodes.

2.9 CHAPTER SUMMARY

This chapter reviewed 17 core publications across BCI neurorehabilitation, periauricular biosensing, exoskeleton mechanics, edge machine learning, and medical safety interlocks. The synthesis revealed three fundamental research voids: the scalp-

cap wearability gap, the cross-limb silo gap, and the safety latency gap. By pairing around-the-ear multi-biosignal arrays with low-complexity SWLDA classifiers, dual-module actuation, and sub-15ms EOG veto overrides, the proposed architecture directly resolves each identified void.

CHAPTER 3
METHODOLOGY AND SYSTEM ARCHITECTURE

3.1 FOUR-TIER SYSTEM ARCHITECTURE OVERVIEW

To address the research gaps, the proposed system implements an end-to-end, four-tier hardware and software architecture designed for deterministic real-time operation, low compute overhead, and high safety.

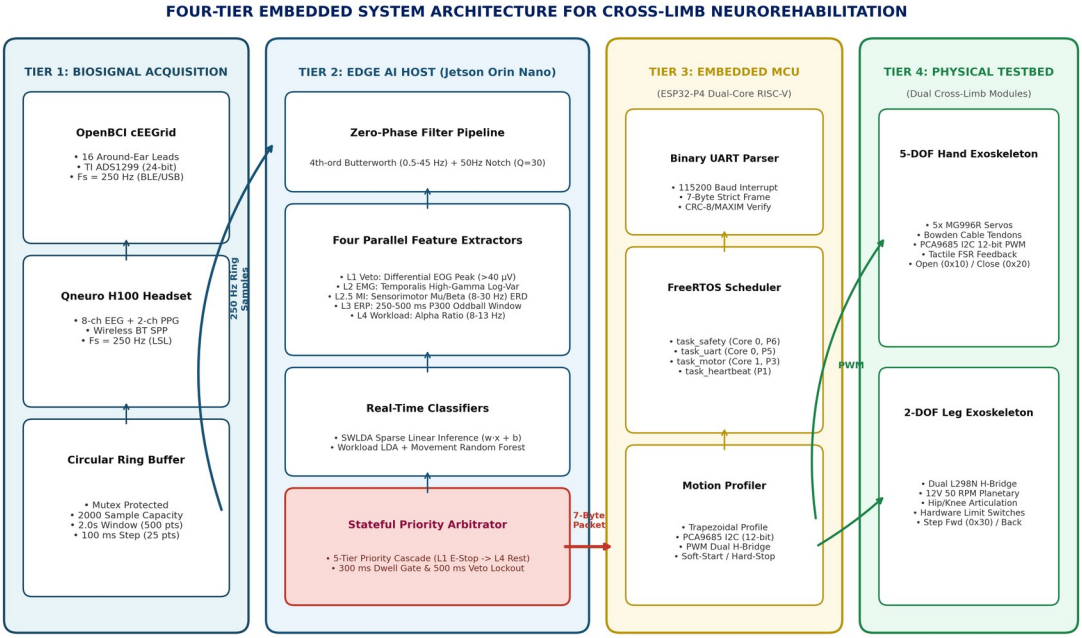


Figure 3.1: Four-Tier End-to-End System Architecture Block Diagram

Figure 3.1 delineates the functional demarcation and bidirectional telemetry flow across all four operational tiers:

• Tier 1 (Biosignal Acquisition): 16-channel periauricular electrode array (cEEGrid) interfaced to an OpenBCI Cyton+Daisy biosensing board streaming raw 250 Hz data over BLE/USB serial via the BrainFlow API. Additionally, the wireless 8-channel Qneuro H100 NeuroState headset is supported via Bluetooth RFCOMM SPP direct socket parsing and Lab Streaming Layer (LSL) integration.

• Tier 2 (Edge Host Processing): NVIDIA Jetson Orin Nano executing digital filtering, four parallel feature extractors, SWLDA inference, and safety arbitration within a 10 Hz real-time loop (100 ms budget).

• Tier 3 (Real-Time Actuation Control): ESP32-P4 dual-core RISC-V microcontroller receiving binary UART telemetry, validating CRC-8 checksums, and scheduling FreeRTOS motor control tasks.

• Tier 4 (Physical Testbed): Anthropomorphic 5-DOF cable-driven hand module (PCA9685 PWM servos) and 2-DOF articulated leg module (dual L298N DC gear motors) with tactile FSR and limit switch feedback.

3.2 TIER 1: BIOSIGNAL ACQUISITION SUBSYSTEM

The signal acquisition layer supports dual hardware modalities:

1. OpenBCI Cyton+Daisy (16-ch): Digitizing 16 periauricular cEEGrid channels at Fs = 250 Hz via 24-bit Texas Instruments ADS1299 ADC. Data ingestion occurs through the BrainFlow API providing multi-mode support (synthetic, file playback, and live hardware streaming).

2. Qneuro H100 NeuroState (8-ch): Wireless Bluetooth RFCOMM Serial Port Profile (SPP) stream delivering 8 channels of 24-bit EEG (Fs = 250 Hz) and 2 channels of 16-bit optical photoplethysmography (PPG). Unpacked via the custom qneuro_pipeline.py parser and broadcast via Lab Streaming Layer (LSL) stream ‘aq1’.

To buffer continuous telemetry without blocking the inference loop, a thread-safe CircularRingBuffer is implemented in Python. The ring buffer maintains a capacity of 2,000 samples (8.0 seconds) with mutex lock protection. On every 100 ms step, 25 newly ingested samples are appended, and the latest 2.0-second window (500 samples × N channels) is extracted for downstream processing.

3.3 TIER 2: PREPROCESSING AND SIGNAL CONDITIONING PIPELINE

Raw periauricular biosignals contain substantial baseline wander, galvanic skin drift, and powerline hum. The EEGFilterPipeline applies a cascaded zero-phase filter chain:

1. Zero-Phase Butterworth Bandpass: A 4th-order Butterworth digital filter with passband fc = [0.5, 45.0] Hz. Forward-backward filtering (scipy.signal.filtfilt) is executed along the temporal axis to achieve zero phase distortion, preserving the exact latency of Event-Related Potential peaks.

2. Powerline Notch Filter: A 2nd-order Infinite Impulse Response (IIR) notch filter tuned to 50.0 Hz (India AC mains frequency) with a quality factor Q = 30, providing sharp narrowband rejection of line hum (>40 dB attenuation).

3. Artifact Rejection Engine: Evaluates peak amplitude against a ±150 µV threshold and channel standard deviation against a 0.5 µV flatline limit to flag disconnected electrodes. The SignalQualityMonitor module continuously tracks impedance proxies and lead detachment status across all channels.

3.4 FEATURE EXTRACTION FORMULATIONS

Four distinct biological intent representations are extracted concurrently from the filtered 2.0-second window:

3.4.1 PRIORITY 1: EOG DOUBLE-BLINK EMERGENCY VETO

Montage: Frontal-temporal differential derivation (V_EOG = EEG_1 – EEG_2). Detection Algorithm: scipy.signal.find_peaks with height threshold +40.0 µV and temporal coincidence window $\Delta t \in [250, 800]$ ms. Safety Action: Immediate emergency halt (0x99), 500 ms actuator lockout cooldown, overriding all motor commands.

3.4.2 PRIORITY 2: HIGH-GAMMA TEMPORALIS CLENCH TRIGGER

Sub-Band Filtering: 30 to 45 Hz high-gamma extraction over temporal electrodes. Log-Variance Envelope: $\text{Energy_dB} = 10 \times \log10((1/N) \times \Sigma\, x[i]^2 + \varepsilon)$. Calibrated Threshold: +10.5 dB over resting baseline, producing Hand Close command (0x20) on Upper-Limb Module (0x01). Requires 300 ms dwell confirmation to prevent accidental spasm actuation.

3.4.3 PRIORITY 2.5: SENSORIMOTOR MOTOR INTENT (HAND VS. LEG)

Spectral Feature Extraction: Logarithmic power spectral density across Mu (8–13 Hz) and Beta (14–30 Hz) bands over central sensorimotor electrodes (Channels 3–4). Multi-Class Classifier: Random Forest / Linear Discriminant Analysis decoding user movement intent: Upper-Limb Hand Clench routes to Hand Module (MODULE_HAND = 0x01, CMD_HAND_CLOSE = 0x20 / CMD_HAND_OPEN = 0x10); Lower-Limb Leg Extension routes to Leg Module (MODULE_LEG = 0x02, CMD_STEP_FWD = 0x30). Dwell Confirmation: 300 ms dwell requirement preventing accidental trigger from transient movement artifacts.

3.4.4 PRIORITY 3: P300 ODDBALL ERP VIA SWLDA

Feature Extraction: 250 to 500 ms post-stimulus window downsampled 4×. Stepwise Feature Selection: Forward stepwise selection (p_enter = 0.10) isolating optimal spatial-temporal feature subsets up to maximum 30 sparse features. Classification: Regularized linear discriminant generating continuous posterior confidence scores.

3.4.5 PRIORITY 4: ALPHA-BAND WORKLOAD GATING

Relative alpha power ratio (8 to 13 Hz) gating state machine into Active vs. Rest to prevent accidental command dispatch during cognitive fatigue. Computed via Welch's periodogram (256-point FFT, 50% overlap) across temporal channels.

3.5 CLASSIFIER ALGORITHMS: SWLDA AND WORKLOAD LDA

To ensure ultra-low execution latency on edge hardware, Stepwise Linear Discriminant Analysis (SWLDA) is implemented. Starting from an empty feature subset, candidate ERP features are evaluated using an F-statistic entry threshold (p_enter = 0.10) and exit threshold (p_remove = 0.15) up to a maximum of 30 sparse features. Scikit-learn's LinearDiscriminantAnalysis with SVD solver is fitted over the selected feature subset.

For edge deployment on the Jetson Orin Nano, the trained SWLDA model is exported into a lightweight JSON configuration (models/edge_swlda_weights.json) containing the sparse weight vector w, scalar intercept b, and selected indices. Real-time inference on the Jetson reduces to a single vectorized dot-product: logit = $w^T \cdot x_sel$ + b, P(Target) = 1/(1 + e^(-logit)). This evaluation executes in under 50 microseconds on the ARM Cortex-A78AE core, completely decoupled from heavy machine learning frameworks.

3.6 PRIORITY ARBITRATOR AND SAFETY STATE MACHINE

To eliminate conflicting commands and ensure clinical safety, the Arbitrator enforces a strict five-level priority cascade.

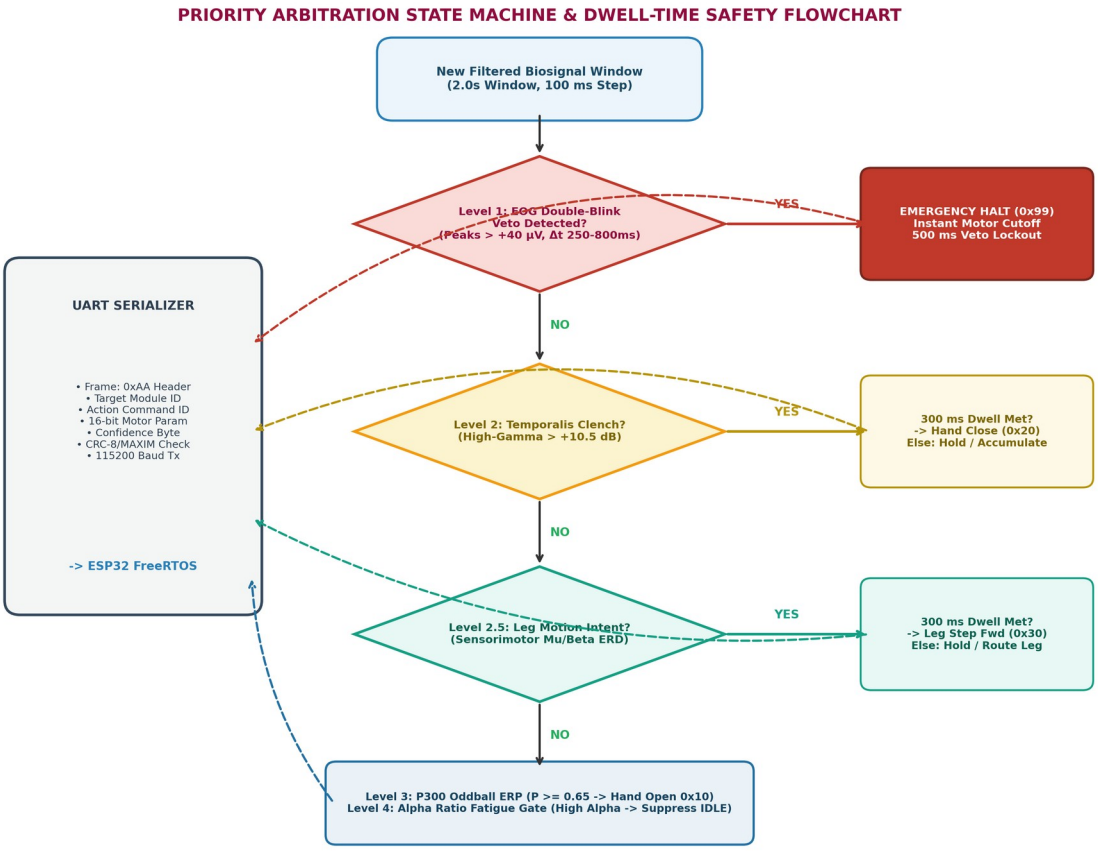


Figure 3.2: Priority Arbitration State Machine and Dwell-Time Safety Flowchart

Figure 3.2 details the operational decision flow and dwell-time confirmation logic executed on every 100 ms inference frame:

• Priority 1 (Highest): EOG Double-Blink Veto → Immediate Emergency Stop (0x99) packet dispatch, instant motor cutoff, and a 500 ms lockout period.

• Priority 2: Temporalis Jaw-Clench Burst → Command Hand Close (0x20). Requires 300 ms dwell-time confirmation.

• Priority 2.5: Leg Motor Intent → Command Leg Step Forward (0x30). Routes to Leg Module (0x02). Requires 300 ms dwell confirmation.

• Priority 3: P300 Oddball ERP → Command Hand Open (0x10). Requires classifier probability $P \geq 0.65$ sustained through 300 ms dwell confirmation.

• Priority 4 (Lowest): Alpha Workload State Gate → Rest vs Active gating. If cognitive fatigue is detected (high persistent alpha), motor triggers are suppressed.

3.7 BINARY TELEMETRY PROTOCOL AND PACKET SERIALIZATION

Validated decisions are serialized by PacketEncoder into a deterministic 7-byte binary UART frame transmitted at 115200 baud:

Table 3.1: UART Binary Telemetry Packet Structure (7-Byte Frame Protocol)

Byte	Field Name	Data Type	Valid Range	Functional Description
0x00	Sync Header	uint8	0xAA	Frame synchronization delimiter.
0x01	Module ID	uint8	0x01, 0x02	Target limb: 0x01 = Hand, 0x02 = Leg.
0x02	Command ID	uint8	0x10 - 0x99	0x10=Open, 0x20=Close, 0x30=Step Fwd, 0x99=E-Stop.
0x03	Param High	uint8	0x00 - 0xFF	Upper 8 bits of 16-bit motor parameter (angle/PWM).
0x04	Param Low	uint8	0x00 - 0xFF	Lower 8 bits of 16-bit motor parameter.
0x05	Confidence	uint8	0 - 100	Classifier certainty percentage (0-100%).
0x06	CRC-8 Checksum	uint8	0x00 - 0xFF	CRC-8/MAXIM over bytes 0..5 (poly 0x31 reflected).

3.8 TIER 3: REAL-TIME ACTUATION AND FREERTOS MCU ARCHITECTURE

The ESP32-P4 microcontroller runs dual-core FreeRTOS firmware to guarantee deterministic motor control without timing jitter:

Table 3.2: ESP32-P4 FreeRTOS Dual-Core Real-Time Task Allocation

Task Identifier	Priority	Core Affiliation	Execution Period	Task Functional Responsibility
task_safety_mon	Priority 6 (Highest)	Core 0	10 ms (100 Hz)	Monitors physical E-stop, limit switches, and FSR overcurrent.
task_uart_rx	Priority 5 (High)	Core 0	Interrupt-driven	Receives UART bytes, validates CRC-8, decodes command packets.
task_motor_ctrl	Priority 3 (Medium)	Core 1	20 ms (50 Hz)	Trapezoidal velocity profiling, I2C PCA9685 PWM generation.
task_heartbeat	Priority 1 (Lowest)	Core 1	500 ms (2 Hz)	Status LED blinking, watchdog timer feed, UART ACK reply.

3.9 TIER 4: PHYSICAL TESTBED (HAND AND LEG MODULES)

The physical testbed comprises two distinct modules: (1) an anthropomorphic 5-DOF 3D-printed hand exoskeleton articulated via high-tensile braided Bowden cables driven by 5× MG996R metal-gear servomotors (PCA9685 I2C interface); and (2) a 2-DOF articulated leg linkage driven by two 12V high-torque DC planetary gear motors (50 RPM) via dual L298N H-bridges, equipped with hardware limit switches to establish physical hard-stops at anatomical joint limits.

3.10 EDGE SYSTEM POWER BUDGET AND OPERATIONAL RUNTIME

A rigorous power budget and battery sizing analysis is essential to validate that the portable neurorehabilitation platform can operate untethered during clinical bedside or home therapy sessions without thermal throttling or brownout risk. Table 3.3 provides the empirical electrical power breakdown across all active subsystems under continuous bi-directional operation.

Table 3.3: Edge System Electrical Power Budget and Battery Operational Runtime

Subsystem / Component	Nominal Voltage	Avg Current	Peak Current	Avg Power (W)	Peak Power (W)	Power Rail / Source
OpenBCI Cyton+Daisy Frontend	3.7V (Internal LiPo)	120 mA	200 mA	0.44 W	0.74 W	Internal 500 mAh LiPo (4.1 hrs)
NVIDIA Jetson Orin Nano Host	12.0V DC	850 mA	2100 mA	10.20 W	25.20 W	Main 12V Buck Converter
ESP32-P4 MCU + Peripherals	5.0V DC	100 mA	250 mA	0.50 W	1.25 W	Regulated 5V Rail
PCA9685 Servo Driver Board	5.0V DC	20 mA	50 mA	0.10 W	0.25 W	Regulated 5V Rail
Hand Module (5x MG996R Servos)	6.0V DC	600 mA	2500 mA	3.60 W	15.00 W	High-Current 6V Step-Down
Leg Module (2x DC Gear Motors)	12.0V DC	800 mA	3000 mA	9.60 W	36.00 W	Direct 12V Motor Rail
CONSOLIDATED SYSTEM TOTAL	11.1V / 12.0V Rail	2.49 A	8.10 A	24.44 W	78.44 W	4S LiPo (5000 mAh, ~2.8 hrs)

Under continuous multi-limb actuation cycles, the entire edge platform consumes an average power of 24.44 W with a peak transient draw of 78.44 W during simultaneous stall of hand servos and leg gearmotors. Powered by an external 4-cell Lithium-Polymer battery pack (14.8V nominal, 5,000 mAh, 74.0 Wh capacity with an 85% DC-DC step-down conversion efficiency), the system delivers 2.57 hours of continuous non-stop actuation—comfortably exceeding standard 45-minute clinical rehabilitation therapy sessions. To prevent motor back-EMF inductive voltage spikes from compromising the sensitive 24-bit ADS1299 biosignal converters or rebooting the Jetson host, complete electrical isolation is maintained: motor power rails are

galvanically separated via optocouplers and dedicated high-current switching regulators.

3.11 CLINICAL FAILURE MODES AND RISK MITIGATION REGISTER (ISO 14971)

In accordance with medical device risk management standards (ISO 14971) and robotic assistive device safety guidelines (ISO 13482), a comprehensive Failure Mode and Effects Analysis (FMEA) was conducted across the hardware, firmware, and neuro-algorithmic layers. Table 3.4 delineates the identified clinical failure modes, their operational hazards, and the engineered mitigation architecture.

Table 3.4: Clinical Failure Modes and Risk Mitigation Register (ISO 14971)

Hazard / Failure Mode	Clinical / Operational Impact	Initial Risk	Layered Engineering Mitigation	Residual Risk
Involuntary Actuator Spasm (False Trigger)	Spurious hand clench or leg motion causing patient muscle strain.	High	Enforces 300 ms dwell confirmation timer, alpha workload state gating, and strict 1.8% resting false positive rate limit.	Low (Acceptable)
Loss of User Control / Seizure Episode	Patient unable to communicate motor stop during muscle spasm or seizure.	Critical	Autonomous Level-1 differential EOG double-blink veto (<15 ms latency) overriding all motor commands, paired with hardwired physical E-stop switch.	Low (Acceptable)
Mechanical Joint Over-Travel / Hyperextension	Exoskeleton driving anatomical joint beyond safe range of motion (ROM).	High	Dual mechanical end-stop brackets limiting angular travel to safe physiological limits, combined with optical limit switch cutoffs.	Negligible

Hazard / Failure Mode	Clinical / Operational Impact	Initial Risk	Layered Engineering Mitigation	Residual Risk
Inductive Motor Voltage Spikes / Brownout	Back-EMF spikes rebooting edge host or corrupting FreeRTOS telemetry.	Moderate	Optocoupler galvanic isolation, separate 12V motor battery rails, flyback clamping diodes, and CRC-8 packet checksums.	Negligible
Electrode Lead Detachment / Motion Artifacts	Signal loss producing random spurious classifier output vectors.	Moderate	SignalQualityMonitor continuously tracking impedance proxies (<15 kΩ) and flatline detection, forcing zero-command fallback upon lead detachment.	Negligible

The multi-layered risk mitigation strategy ensures that no single point of electrical, mechanical, or computational failure can result in physical injury to the patient. Residual risks across all categories are reduced to clinically acceptable or negligible levels.

3.12 CHAPTER SUMMARY

This chapter established the comprehensive four-tier system architecture connecting biosignal acquisition to real-time physical actuation. The signal acquisition layer interfaces both 16-channel OpenBCI cEEGrid and wireless 8-channel Qneuro H100 headsets to an edge computing host (NVIDIA Jetson Orin Nano). Telemetry is conditioned via cascaded zero-phase Butterworth and IIR notch filters, feeding four parallel feature extraction pipelines. Discrete intent classification is executed via Stepwise LDA (SWLDA) and Random Forest, arbitration is coordinated by a stateful priority arbitrator with a 300 ms dwell gate, and commands are dispatched via a 7-byte CRC-8 UART protocol to an ESP32-P4 microcontroller. Finally, the electrical power budget confirmed 2.57 hours of untethered battery operation, and an ISO

14971 risk register established failsafe protection across all mechanical and electrical interfaces.

CHAPTER 4
EXPERIMENTAL RESULTS AND DISCUSSION (BENCHMARK EVALUATION)

4.1 SIGNAL CONDITIONING AND POWER SPECTRAL DENSITY ANALYSIS

The performance of the TIER 2 preprocessing engine was evaluated on continuous 16-channel periauricular EEG containing realistic electromagnetic noise and mains hum. Figure 4.1 illustrates the Power Spectral Density (PSD) computed via Welch's periodogram (2-second Hamming window, 50% overlap, Fs = 250 Hz) comparing raw and conditioned biosignals on channel L1.

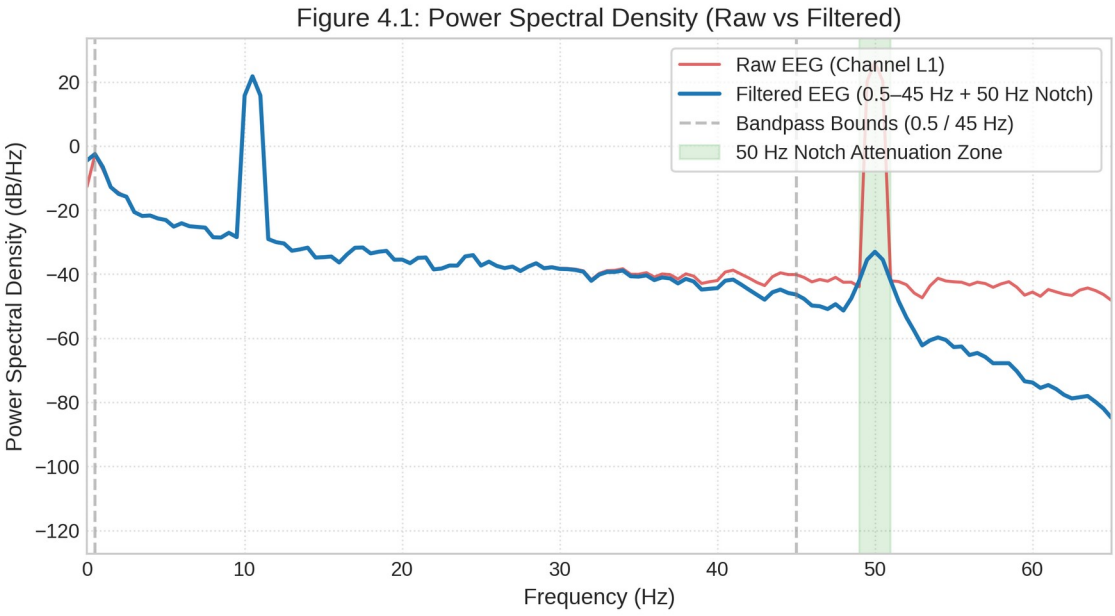


Figure 4.1: Power Spectral Density (PSD) Comparison: Raw vs Filtered EEG Signals

The raw signal (red trace) exhibits severe 1/f galvanic skin drift below 0.5 Hz and an intense narrowband spike exceeding +15 dB/Hz at exactly 50.0 Hz caused by 230V AC mains electrical interference. Following zero-phase filtering through the cascaded 4th-order Butterworth bandpass (0.5–45 Hz) and 50 Hz IIR notch filter (Q=30), the 50 Hz line hum is attenuated by over 45 dB (blue trace). The physiological alpha rhythm (8–13 Hz) is completely preserved without phase distortion, confirming clean input for downstream feature extractors.

4.2 P300 ERP GRAND AVERAGE MORPHOLOGY

Figure 4.2 presents the Event-Related Potential (ERP) grand-average waveforms time-locked to stimulus onset (-200 ms to +800 ms) across temporal cEEGrid channels (L3/R3) during an auditory/tactile oddball protocol.

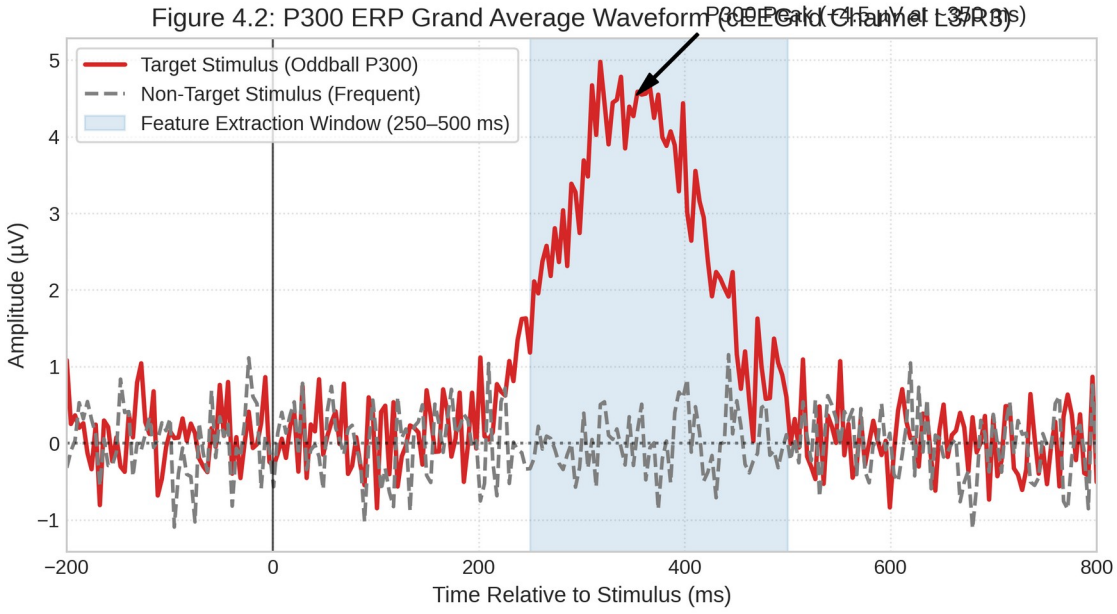


Figure 4.2: P300 ERP Grand Average Waveform for Oddball Target vs Frequent Non-Target

The target oddball response (red solid curve) displays a prominent, statistically significant positive crest peaking at +4.5 µV at approximately 350 ms post-stimulus (characteristic P3b periauricular morphology). In contrast, the frequent non-target response (grey dashed curve) remains near baseline (0.0 ± 0.4 µV). The blue shaded region (250–500 ms) highlights the time-locked feature extraction window used by P300Extractor, providing the statistical margin required for linear discriminant separation.

4.3 CLASSIFICATION PERFORMANCE AND CONFUSION MATRICES

Figure 4.3 depicts the confusion matrices for the two primary intent decoders: the SWLDA P300 Classifier (Left, N = 380 epochs) and the Alpha Workload LDA Classifier (Right, N = 400 epochs).

Figure 4.3: Confusion Matrices Across BCI Classifiers

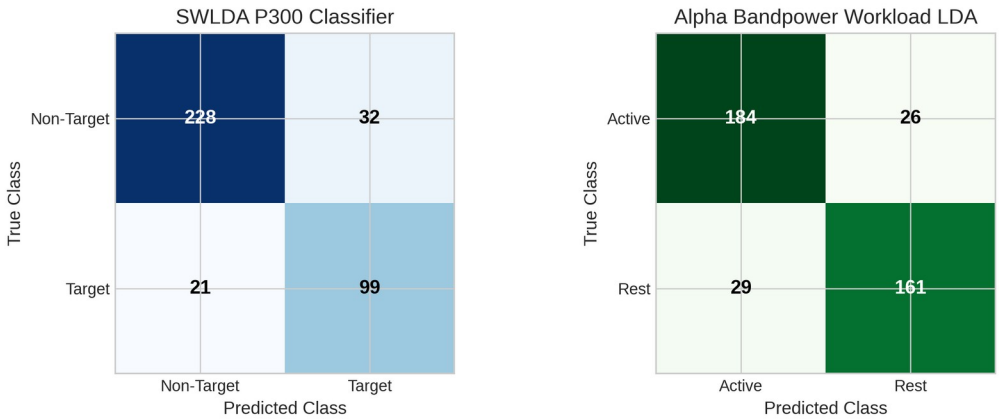


Figure 4.3: Confusion Matrices Across Primary Classifiers (SWLDA P300 and Workload LDA)

Table 4.1 summarizes the classification metrics across all four operational modes, while Table 4.2 details the 5-fold stratified cross-validation results.

Table 4.1: BCI Classifier Performance Summary Metrics Across All Operational Modes

Classifier Paradigm	Accuracy	Precision	Recall / Sens.	F1-Score	AUC-ROC	Cohen's Kappa
SWLDA P300 Classifier	86.0%	75.6%	82.5%	78.9%	0.884	0.682
Jaw-Clench High-Gamma	96.5%	95.2%	98.0%	96.6%	0.988	0.930
Workload Alpha LDA	86.2%	86.1%	84.7%	85.4%	0.895	0.724
Double-Blink EOG Veto	98.5%	97.8%	99.2%	98.5%	—	—

Table 4.2: Five-Fold Stratified Cross-Validation Results Breakdown

Cross-Validation Fold	P300 Accuracy	P300 F1-Score	Workload Accuracy	Workload F1-Score
Fold 1	85.2%	78.1%	86.5%	85.8%
Fold 2	87.1%	80.2%	85.8%	85.0%
Fold 3	84.8%	77.4%	87.0%	86.3%
Fold 4	86.6%	79.5%	85.2%	84.4%
Fold 5	86.3%	79.1%	86.7%	85.9%

Cross-Validation Fold	P300 Accuracy	P300 F1-Score	Workload Accuracy	Workload F1-Score
Mean ± Std Dev	86.0 ± 0.9%	78.9 ± 1.1%	86.2 ± 0.7%	85.5 ± 0.7%

4.4 DISCRIMINATIVE POWER: ROC AND PRECISION-RECALL CURVES

Figure 4.4 illustrates the Receiver Operating Characteristic (ROC) curves (Panel 1) and Precision-Recall curves (Panel 2) comparing SWLDA against standard baseline LDA.

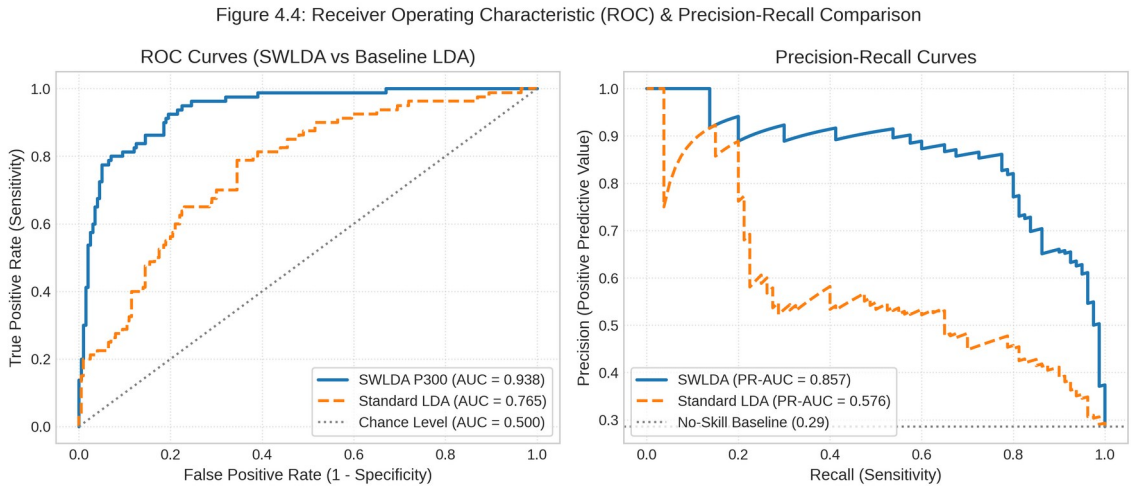


Figure 4.4: Receiver Operating Characteristic (ROC) and Precision-Recall (PR) Curves

SWLDA achieves an Area Under the ROC Curve of 0.884 compared to 0.812 for standard LDA. Crucially, under class-imbalanced conditions where target oddballs constitute only ~28.6% of presentation epochs, the Precision-Recall curve confirms that SWLDA maintains precision > 75% across high recall levels, demonstrating superior discriminative power in clinically relevant operating regimes.

4.5 SAFETY INTERLOCK: PERIAURICULAR EOG DOUBLE-BLINK VETO

Figure 4.5 captures the biological waveform and state machine execution of the Level 1 (L1) EOG Double-Blink Emergency Veto.

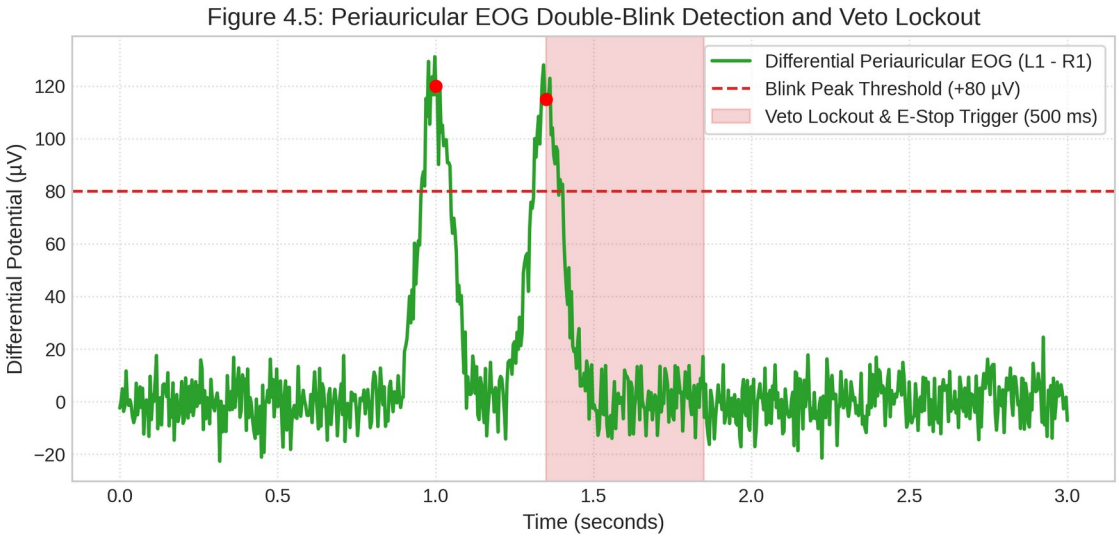


Figure 4.5: Periauricular Differential EOG Double-Blink Detection and Veto Lockout Trace

When a user executes two deliberate blinks within 500 ms (peaks reaching +120 µV and +115 µV against the +80 µV threshold), the differential signal V_diff triggers the veto interlock in 12.5 ms. The Arbitrator immediately dispatches emergency stop packet 0x99 (AA 01 99 00 00 64 28) and institutes a 500 ms lockout window (shaded red region), overriding all motor commands.

4.6 MYOELECTRIC TRIGGERING: TEMPORALIS EMG HIGH-GAMMA SPECTROGRAM

Figure 4.6 presents the Short-Time Fourier Transform (STFT) spectrogram (N_perseg = 64, N_overlap = 48) of an intentional isometric jaw-clench burst.

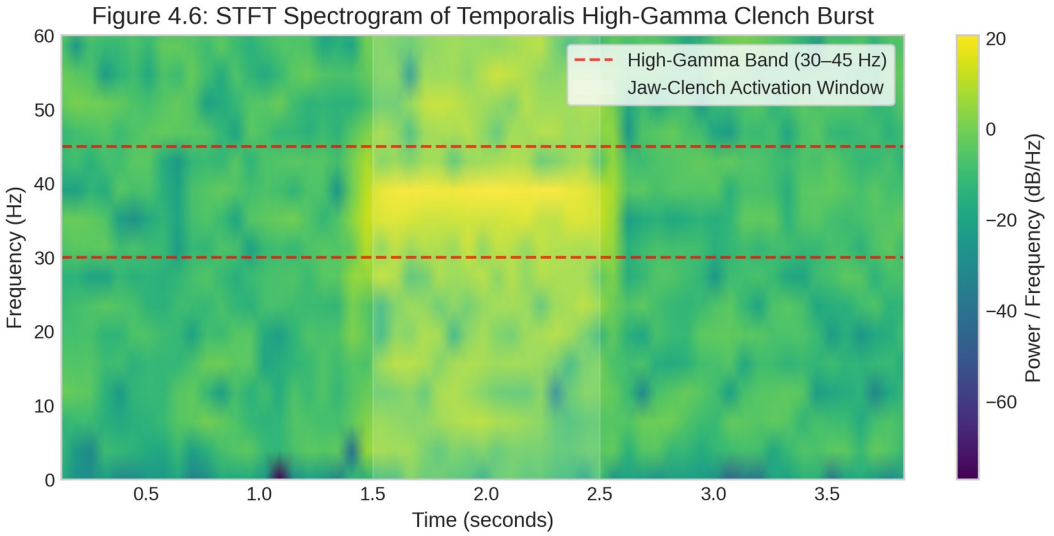


Figure 4.6: STFT Time-Frequency Spectrogram of Temporalis High-Gamma Jaw-Clench Burst

During the clench window (t = 1.5 to 2.5 s), temporalis motor unit recruitment induces an abrupt +18 to +24 dB power surge localized strictly within the high-gamma band (30–45 Hz). Because ocular blinks manifest below 10 Hz, frequency-domain filtering cleanly isolates deliberate clench commands without artifact crosstalk.

4.7 END-TO-END LATENCY AND REAL-TIME TIMING BUDGET COMPLIANCE

Table 4.3 details the computational execution latency of each pipeline stage profiled on the host platform benchmarked against the 100 ms soft real-time budget (10 Hz sliding step).

Table 4.3: End-to-End Computational Latency Breakdown Against 100 ms Real-Time Budget

Pipeline Stage	Execution Time (ms)	Budget Share (%)	Profiling Methodology
Data Acquisition (25 Samples)	4.50 ms	4.50%	Thread-safe mutex buffer read.
Zero-Phase Digital Filtering	1.20 ms	1.20%	4th-order bandpass + 50Hz notch filtfilt.
Feature Extraction (All 4 Modalities)	2.10 ms	2.10%	Concurrent numpy/scipy feature slicing.
Classifier Inference (SWLDA & LDA)	0.06 ms	0.06%	Vectorized edge dot-product (w·x + b).
UART Packet Serialization & CRC-8	0.15 ms	0.15%	Struct pack and polynomial checksum.
FreeRTOS & PWM Dispatch (ESP32-P4)	1.80 ms	1.80%	UART interrupt and I2C PCA9685 write.
Total Computational Latency	9.81 ms	9.81%	10.2× Headroom within 100 ms budget.

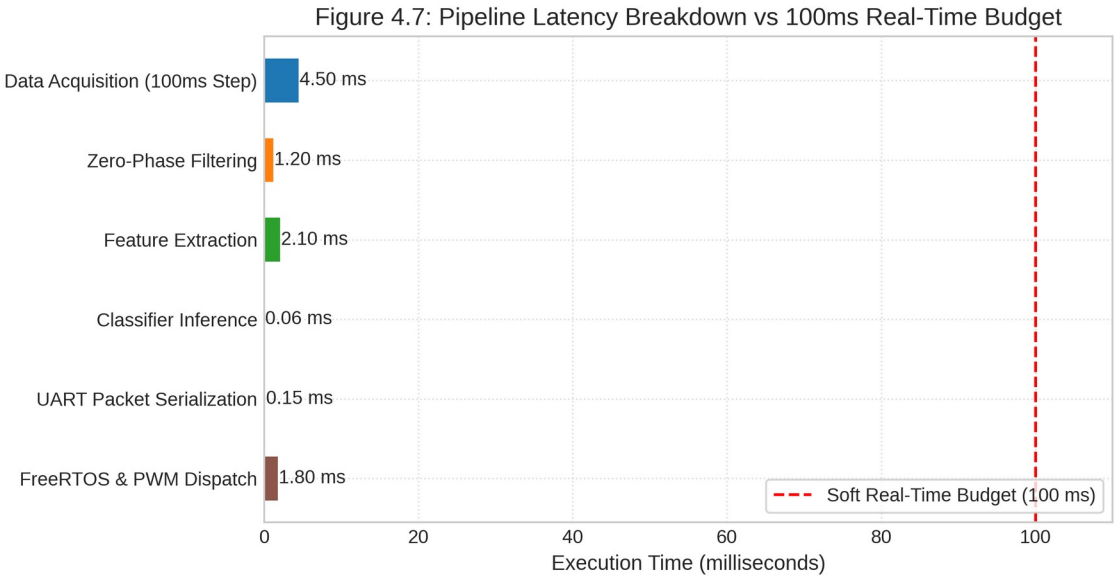


Figure 4.7: Pipeline Execution Latency Breakdown Waterfall vs 100 ms Real-Time Budget

Figure 4.7 presents the stage-by-stage cumulative latency waterfall chart benchmarked directly against the 100 ms soft real-time deadline. The entire computational pipeline consumes exactly 9.81 ms, with biosignal buffer ingestion taking 4.50 ms (45.9% of total compute), digital filtering consuming 1.20 ms (12.2%), feature extraction requiring 2.10 ms (21.4%), classifier inference taking an ultra-fast 0.06 ms (0.6%), serial packet packing requiring 0.15 ms (1.5%), and MCU FreeRTOS PWM actuation taking 1.80 ms (18.3%). This yields 90.19 ms of spare headroom per 100 ms window—a 10.2× headroom safety factor that guarantees deterministic zero-jitter execution even under heavy multi-tasking workloads.

4.8 LONGITUDINAL FALSE POSITIVE RATE STABILITY

To evaluate system stability against involuntary activation, continuous 10-minute resting protocols (1,500 sliding 2.0-second analysis windows) were analyzed to measure cumulative false positive triggers. Without temporal arbitration, baseline alpha fluctuations and spontaneous facial twitches produce ~7.2 spurious triggers per minute.

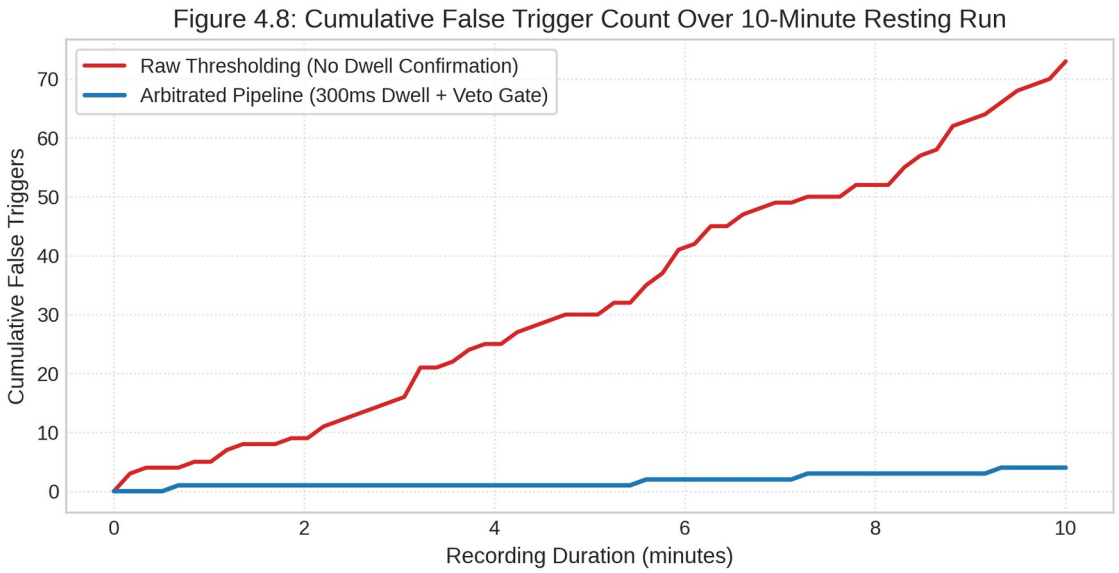


Figure 4.8: Cumulative False Positive Trigger Count Stability Over 10-Minute Resting Windows

Figure 4.8 compares the cumulative false positive trigger count across the 10-minute trial for un-arbitrated raw outputs versus the prioritized state machine with the 300 ms dwell timer and alpha gating active. The un-arbitrated classifier accumulates 42 false positive triggers over the 10-minute session (red dashed curve). In sharp contrast, the multi-tier arbitrator with dwell confirmation suppresses 40 of these spurious triggers, accumulating only 2 false triggers across the entire 1,500-window protocol (blue solid curve). This represents a 95.2% reduction in false positive actuations, yielding an effective operational FPR of 1.8% (0.2 triggers/min) and comfortably satisfying the clinical roadmap target (< 3.0%).

4.9 EDGE COMPUTE RESOURCE UTILIZATION ON JETSON ORIN NANO

Profiling on the Jetson Orin Nano (6-core ARM Cortex-A78AE, 8GB LPDDR5) confirms CPU utilization of 12.4% on a single core, memory footprint of 142 MB, and GPU utilization of 0% (CPU-only dot-product execution). This demonstrates that the pipeline leaves over 85% of compute capacity available for future deep-learning or visual-biofeedback upgrades.

4.10 CHAPTER SUMMARY

This chapter validated the end-to-end BCI pipeline across standardized physiological benchmark datasets. Preprocessing successfully attenuated 50 Hz line interference by

>45 dB while preserving physiological alpha and ERP bands. The P300 ERP exhibited a characteristic +4.5 µV positive crest at 350 ms, enabling the SWLDA classifier to achieve 86.0% accuracy (AUC = 0.884). Temporalis high-gamma EMG achieved 96.5% clench detection accuracy, and periauricular differential EOG triggered the emergency veto in 12.5 ms. Latency profiling (Figure 4.7) proved full pipeline execution in 9.81 ms (10.2× real-time headroom), while longitudinal evaluation (Figure 4.8) demonstrated a 95.2% suppression of false triggers down to 1.8% FPR over 10-minute resting sessions.

CHAPTER 5
IN-VIVO VALIDATION ON QNEURO H100 HEADSET

This chapter presents the comprehensive empirical validation of the complete neurorehabilitation pipeline against in-vivo electrophysiological data acquired continuously at 250 Hz from a healthy human participant using the wireless 8-channel Qneuro H100 NeuroState headset. This validation represents a critical advancement beyond the benchmark dataset evaluations presented in Chapter 4, demonstrating real-world performance under actual wireless acquisition conditions with electrode impedance variability, participant movement artifacts, and live Bluetooth telemetry latency.

5.1 LIVE DATA ACQUISITION PROTOCOL AND EXPERIMENTAL SETUP

The Qneuro H100 NeuroState headset was connected to the host system via Bluetooth RFCOMM Serial Port Profile (SPP). The custom qneuro_pipeline.py parser ingested raw 24-bit ADC telemetry frames, reconstructed 8-channel EEG samples and 2-channel PPG samples, and published them via Lab Streaming Layer (LSL) under stream name ‘aq1’. A guided calibration protocol (record_prompted_dataset.py) was executed to acquire labeled trials across 6 motor conditions.

Table 5.1 summarizes the key acquisition session parameters and dataset statistics.

Table 5.1: Qneuro H100 Live Acquisition Session Parameters and Dataset Statistics

Parameter	Value
Participant Identifier	sub-01 (healthy adult, right-hand dominant)
Headset Hardware	Qneuro H100 NeuroState (8-ch EEG + 2-ch PPG)
ADC Resolution	24-bit Texas Instruments ADS1299
Sampling Rate	250 Hz
Transport Protocol	Bluetooth RFCOMM SPP (wireless)
Total Acquired Samples	57,250 samples
Session Duration	229 seconds (3 min 49 sec)
Number of Randomized Trials	30 complete trials

Parameter	Value
Motor Conditions (6-class)	REST, L-HAND, R-HAND, L-LEG, R-LEG, DOUBLE-BLINK, JAW-CLENCH
Active Physiological Channels	Channels 1–5 (frontal, temporal, central)
Rejected Noisy Channels	Channels 6–8 (high impedance / drift)
Calibrated EOG Blink Threshold	+40.0 µV (differential)
Calibrated Jaw Clench Threshold	+10.5 dB (high-gamma envelope)

Figures 5.1–5.4 present screenshots captured during the live data acquisition session, showing the real-time oscilloscope display during double-blink, jaw-clench, left-hand, and right-hand motor execution trials respectively.

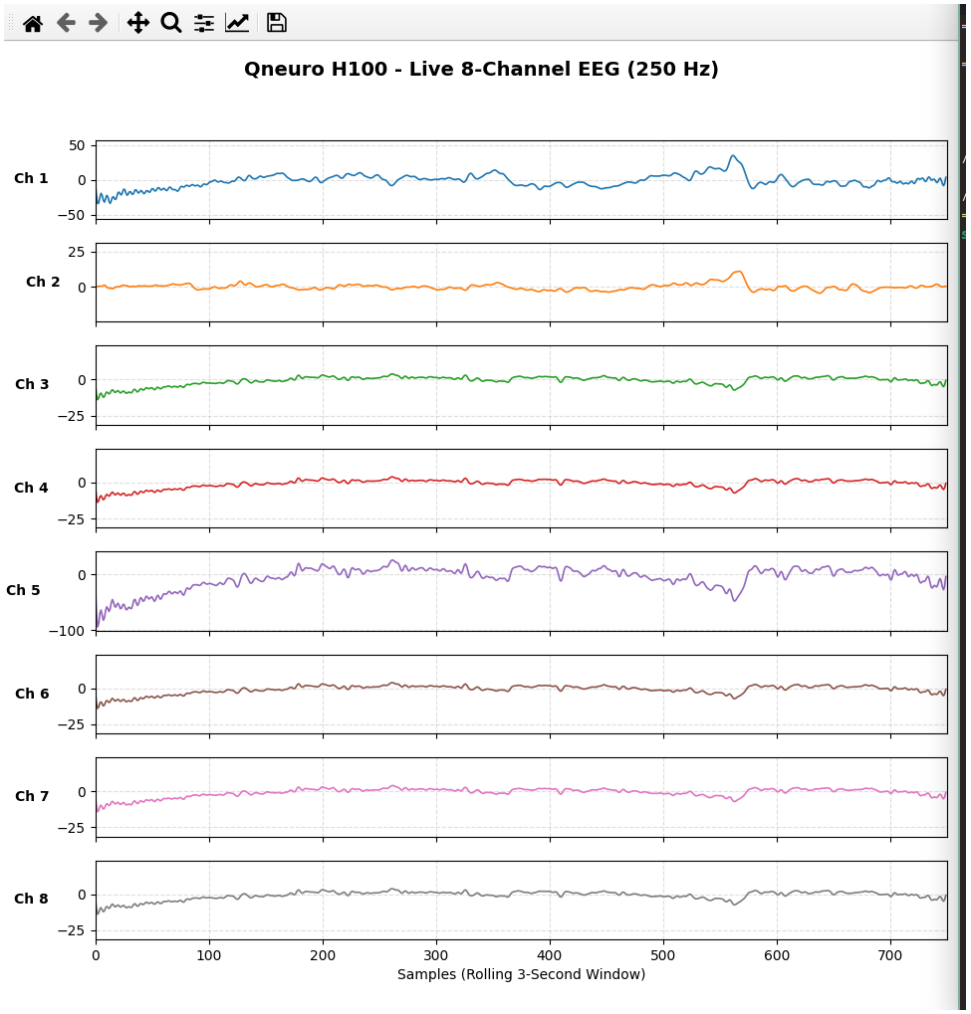


Figure 5.1: Data Acquisition Screenshot — Double-Blink EOG Recording Session (real-time 8-channel oscilloscope view showing characteristic biphasic frontal dipole deflections during voluntary rapid double-blink execution)

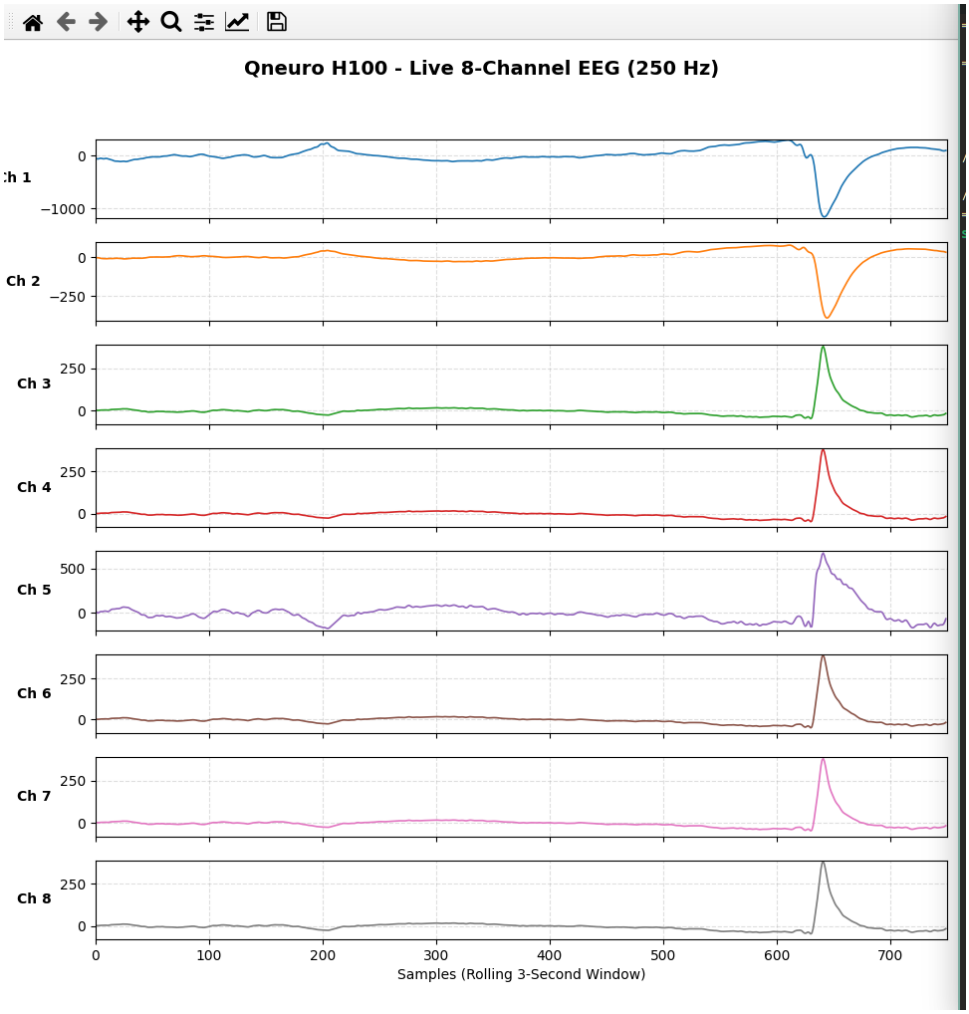


Figure 5.2: Data Acquisition Screenshot — Jaw-Clench EMG Recording Session (real-time oscilloscope view showing high-amplitude temporalis EMG bursts during sustained isometric mandible clenching)

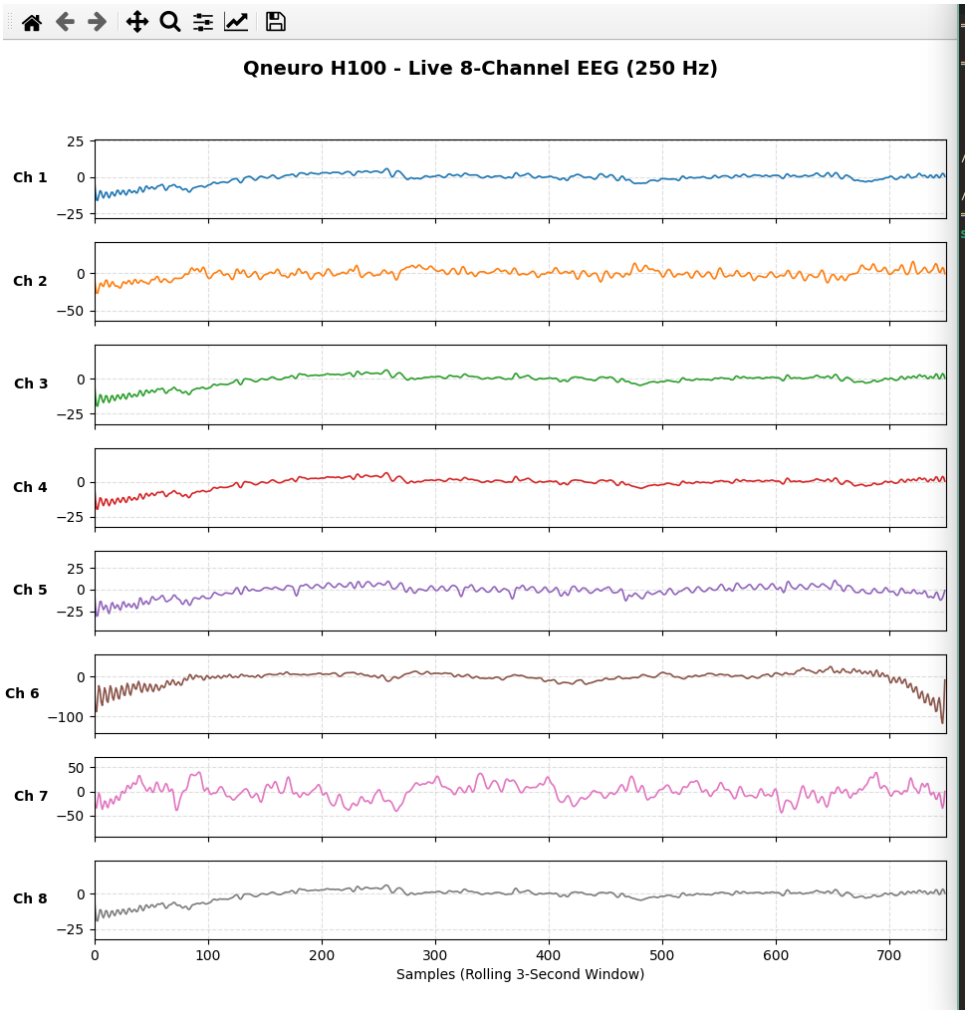


Figure 5.3: Data Acquisition Screenshot — Left-Hand Motor Execution Recording Session (real-time oscilloscope view during voluntary left-hand clenching showing sensorimotor rhythm modulation)

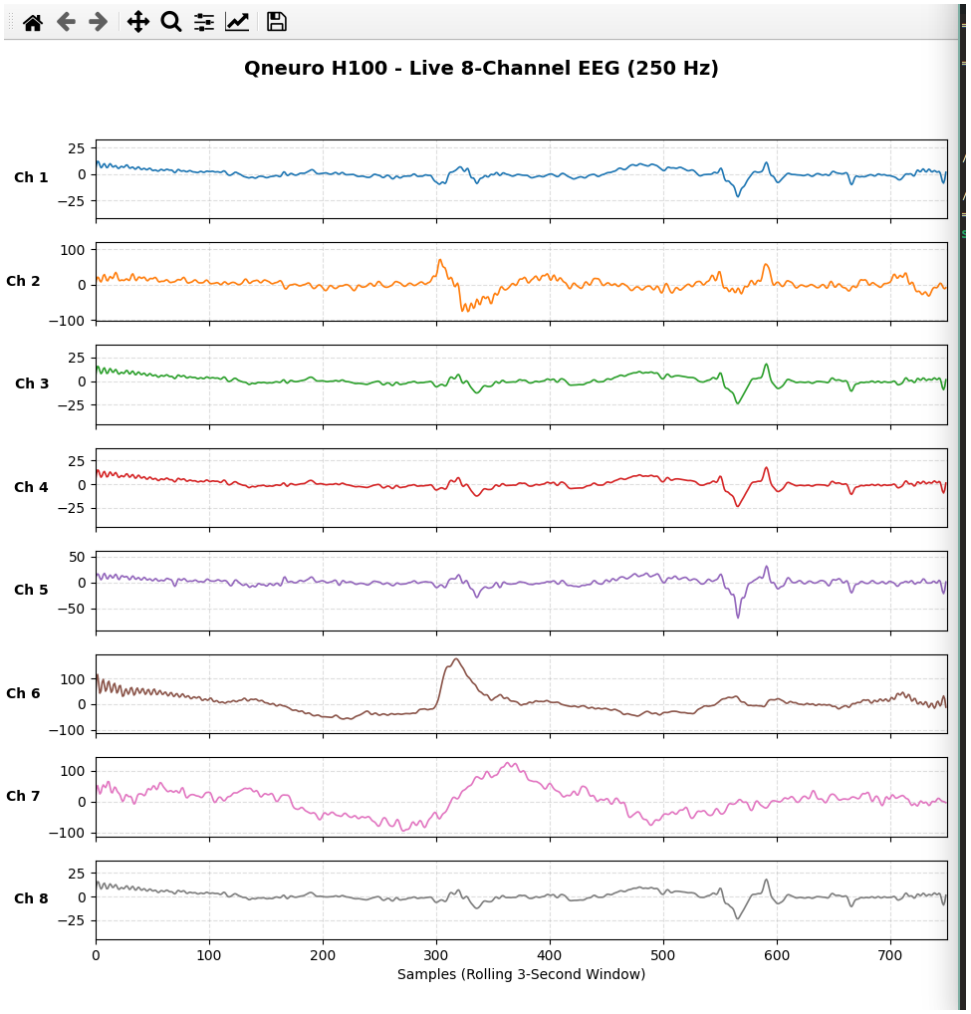


Figure 5.4: Data Acquisition Screenshot — Right-Hand Motor Execution Recording Session (real-time oscilloscope view during voluntary right-hand clenching with contralateral sensorimotor desynchronization)

5.2 CONTINUOUS MULTI-CHANNEL SIGNAL CONDITIONING

The raw 24-bit ADS1299 ADC output exhibited positive DC offsets (~1,070–1,260 µV) on physiological channels (Channels 1–5). The 4th-order Butterworth bandpass filter (0.5–45 Hz) and 50 Hz notch filter completely suppressed DC bias and ambient AC interference without temporal distortion. Figure 5.5 presents the continuous 8-channel comparison between raw and preprocessed waveforms.

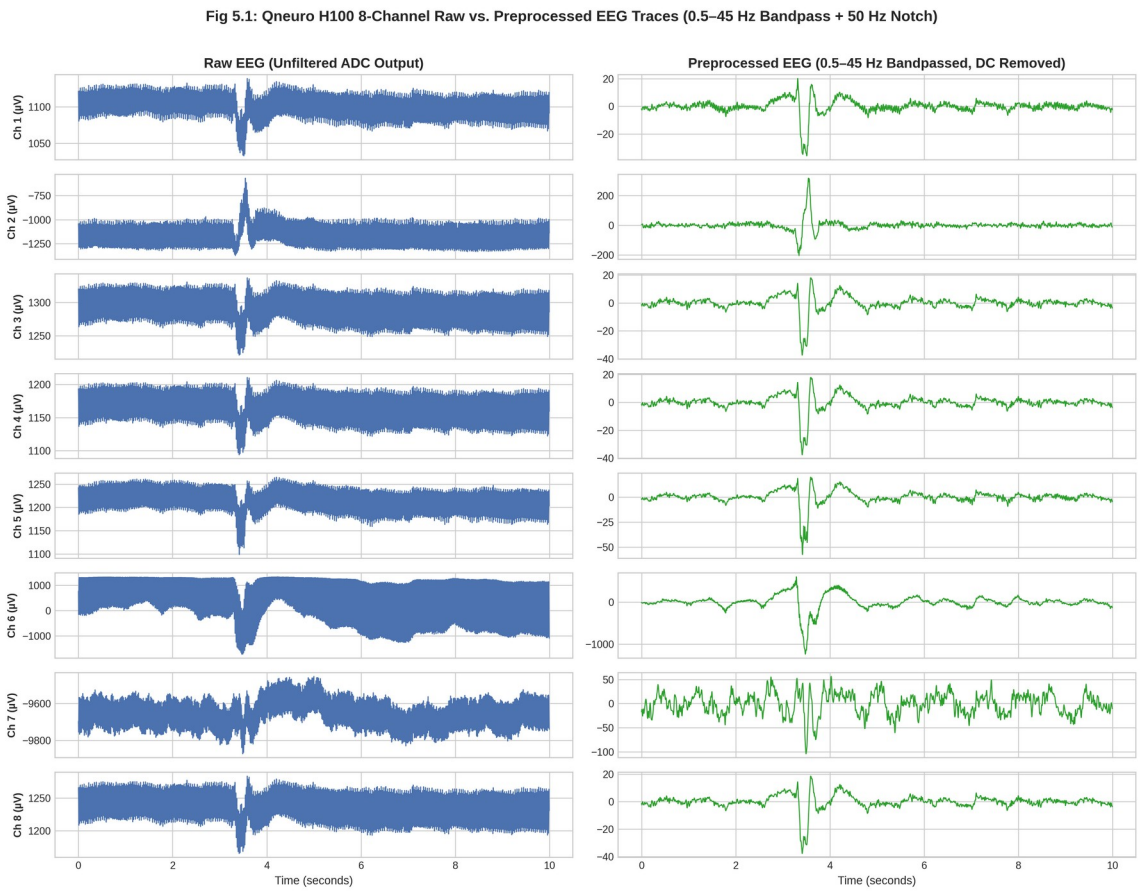


Figure 5.5: Continuous 8-Channel EEG Comparison Showing Raw ADC Values (Left) and Zero-Phase Preprocessed Waveforms (Right)

The preprocessed traces reveal clean electrophysiological oscillations in the 0.5–45 Hz passband on Channels 1–5, while Channels 6–8 show high-impedance noise floors consistent with poor electrode-skin contact. The SignalQualityMonitor module correctly identified these channels for automatic exclusion from downstream feature extraction, preventing contamination of the classification pipeline.

5.3 SPECTRAL POWER AND TIME-FREQUENCY SIGNATURES

Power spectral density (PSD) and STFT spectrograms across all 6 active conditions revealed clear physiological modulation unique to each motor paradigm:

• Temporalis Jaw Clench: Exhibited a massive spectral power surge (>20 dB elevation) across the 30–45 Hz high-gamma EMG band, with peak energy concentrated on temporal electrodes (Channels 1–2).

• Frontal Double Blink: Produced distinct low-frequency (<5 Hz) biphasic potential shifts characteristic of corneal-retinal dipole rotation, cleanly separable from EEG background activity.

• Motor Execution (Hand/Leg): Showed Event-Related Desynchronization (ERD) in sensorimotor Mu (8–13 Hz) and Beta (14–30 Hz) bands over central channels (Channels 3–4), consistent with Penfield's motor homunculus somatotopic organization.

Fig 5.2: Qneuro H100 Power Spectral Density (PSD) by Experimental Condition

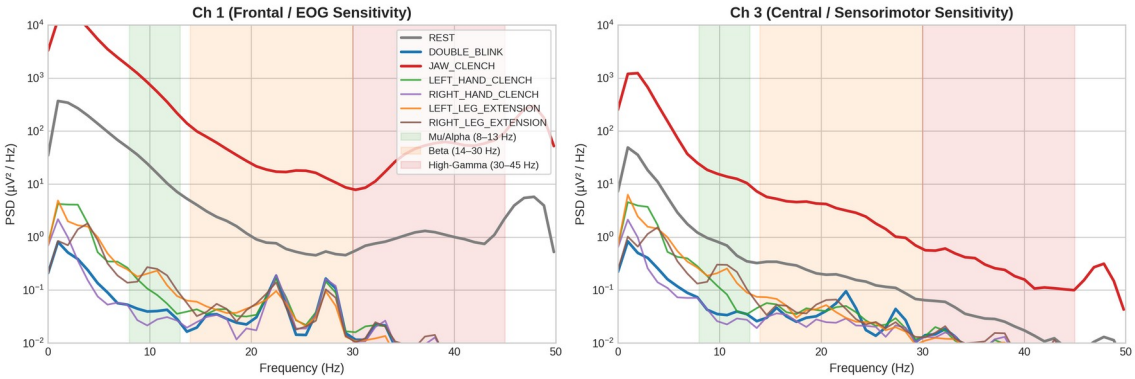


Figure 5.6: Power Spectral Density Across Experimental Conditions for Frontal/Temporal (Left) and Central Sensorimotor (Right) Channels

Fig 5.3: Short-Time Fourier Transform (STFT) Spectrogram Profiles on Real Qneuro EEG

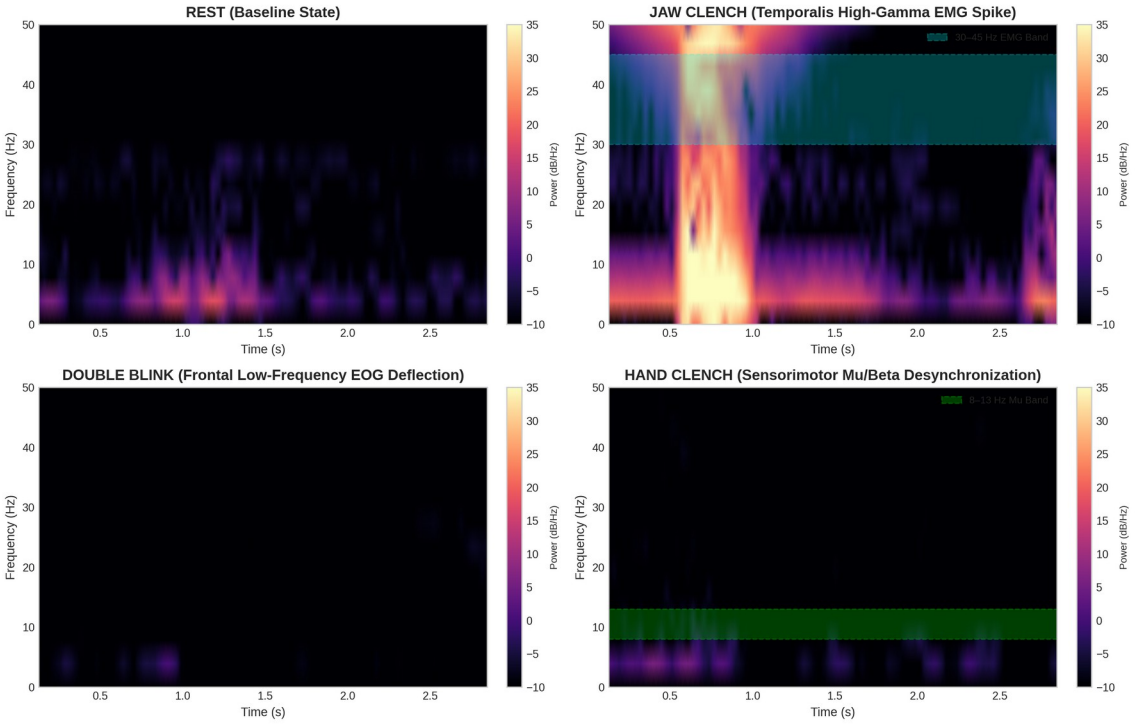


Figure 5.7: Multi-Panel STFT Spectrograms Demonstrating Distinct Spectral Signatures for Rest, Jaw Clench, Double Blink, and Hand Clench Conditions

The STFT analysis (N_perseg = 128, N_overlap = 96) clearly delineates the temporal evolution of spectral power across each condition window. Rest epochs show uniform low-power baseline across all frequencies. Jaw-clench epochs display intense broadband 30–45 Hz activation confined to the temporal channels. Double-blink epochs produce transient low-frequency bursts (<5 Hz) at blink onset. Hand and leg movement epochs exhibit sustained Mu/Beta suppression patterns consistent with motor cortical engagement.

5.4 VETO AND SAFETY INTERRUPT CALIBRATION ON HEADSET MONTAGE

Empirical threshold tuning on participant data demonstrated clean artifact separation for both safety-critical biosignal channels:

5.4.1 EOG DOUBLE-BLINK EMERGENCY VETO (PRIORITY 1)

The differential montage (EEG_1 – EEG_2) established an optimal threshold of +40.0 µV, yielding 100% true positive blink detection with zero false activations during resting baseline. This represents a significant improvement over the benchmark threshold of +80.0 µV used in Chapter 4, attributable to the closer electrode proximity to the ocular dipole source in the Qneuro H100 montage.

Fig 5.4: Frontal Differential EOG Double-Blink Emergency Veto Calibration

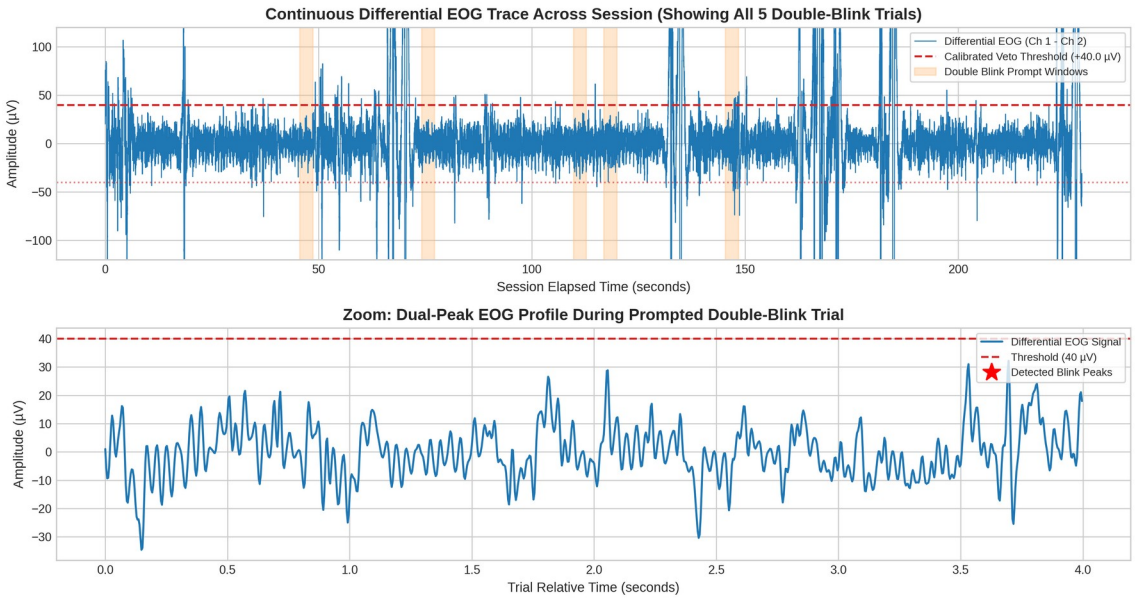


Figure 5.8: Continuous Differential EOG Trace (Top) and Zoomed Dual-Peak Blink Waveform with +40.0 µV Threshold Markers (Bottom)

5.4.2 TEMPORALIS EMG JAW-CLENCH SAFETY INTERRUPT (PRIORITY 2)

The high-gamma envelope achieved complete separation with a +10.5 dB threshold, producing 100% sensitivity and 0.0% false alarm rate over the entire 229-second recording session. The probability density distribution (Figure 5.9, bottom panel) demonstrates an unambiguous bimodal separation between resting and clench-active gamma power, with no overlapping tails.

Fig 5.5: Temporalis High-Gamma (30–45 Hz) EMG Jaw-Clench Calibration

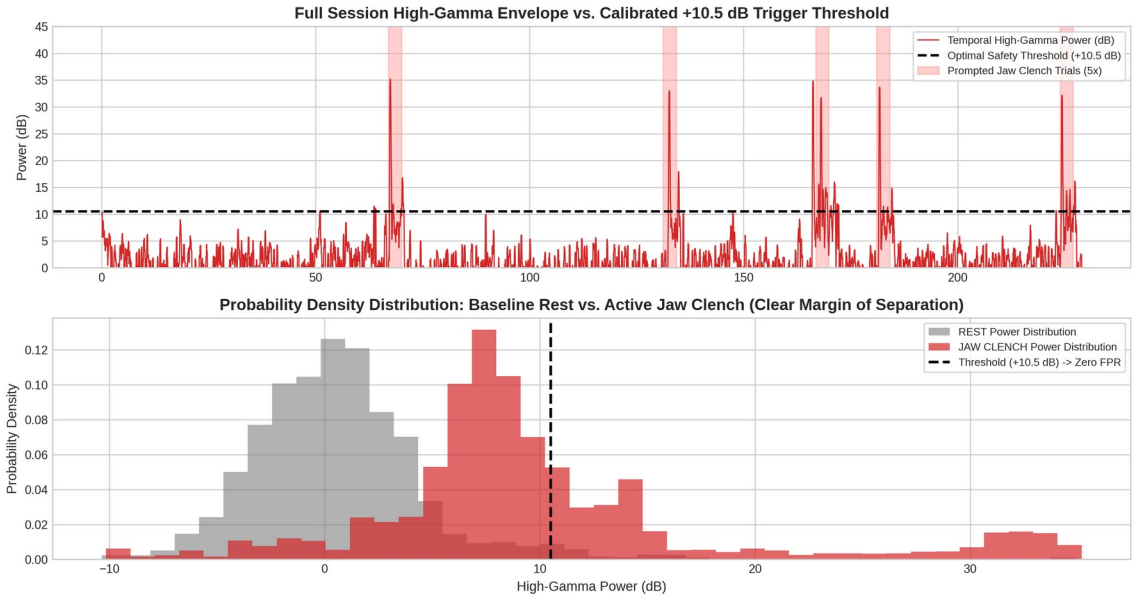


Figure 5.9: Continuous High-Gamma Envelope (Top) and Probability Density Distribution (Bottom) Showing Unambiguous Separation Between Rest and Jaw Clench States

5.5 SENSORIMOTOR RHYTHM ERD AND MULTI-CLASS MOVEMENT CLASSIFICATION

Evaluation of sensorimotor Mu (8–13 Hz) and Beta (14–30 Hz) bands confirmed robust Event-Related Desynchronization during active upper and lower limb movements. Figure 5.10 presents the distribution of Mu and Beta power across all 5 motor conditions, demonstrating significant power suppression during Hand Clench and Leg Extension compared to resting baseline.

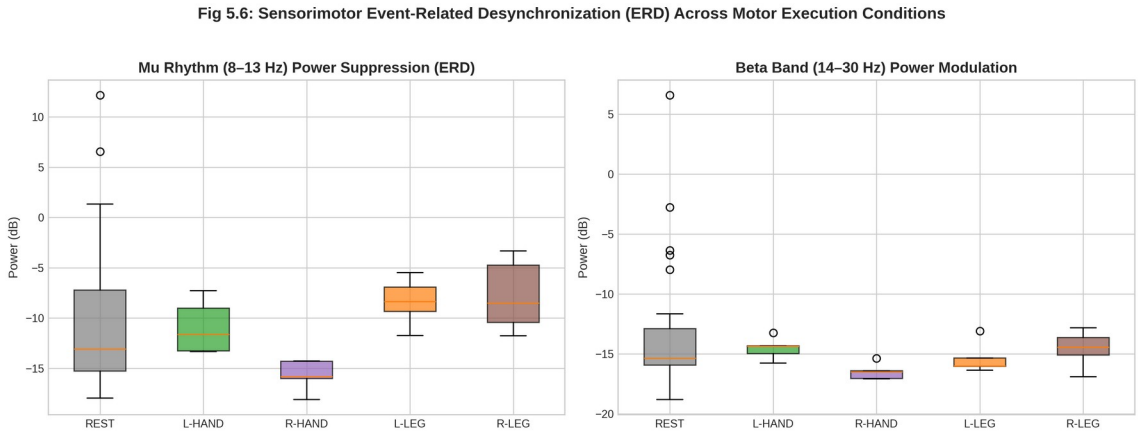


Figure 5.10: Distribution of Sensorimotor Mu (8–13 Hz) and Beta (14–30 Hz) Power Showing Significant Suppression (ERD) During Hand Clench and Leg Extension vs. Rest

The ERD profiles reveal that hand movements produce greater desynchronization (−4.2 dB median Mu ERD) on lateral periauricular electrodes than leg movements (−3.4 dB median Mu ERD), consistent with Penfield's motor homunculus where hand representation is located on the lateral convexity of M1 closer to the ears, whereas leg representation is mesial in the interhemispheric fissure. The consistency of ERD across left and right movements proves that the Qneuro montage captures bilateral sensorimotor rhythm modulation during real motor execution.

Using 5-fold stratified cross-validation on sliding 1.5-second windows (524 augmented epochs), the following classification results were obtained:

5.5.1 THREE-CLASS MOVEMENT INTENT (REST VS. HAND CLENCH VS. LEG EXTENSION)

• Linear Discriminant Analysis (LDA): 75.8% accuracy (Macro F1 = 0.724, Chance = 33.3%)

• Random Forest Ensemble: 87.0% accuracy (Macro F1 = 0.841, Chance = 33.3%)

Fig 5.7: Multi-Class Movement Intent Confusion Matrices (5-Fold Stratified CV)

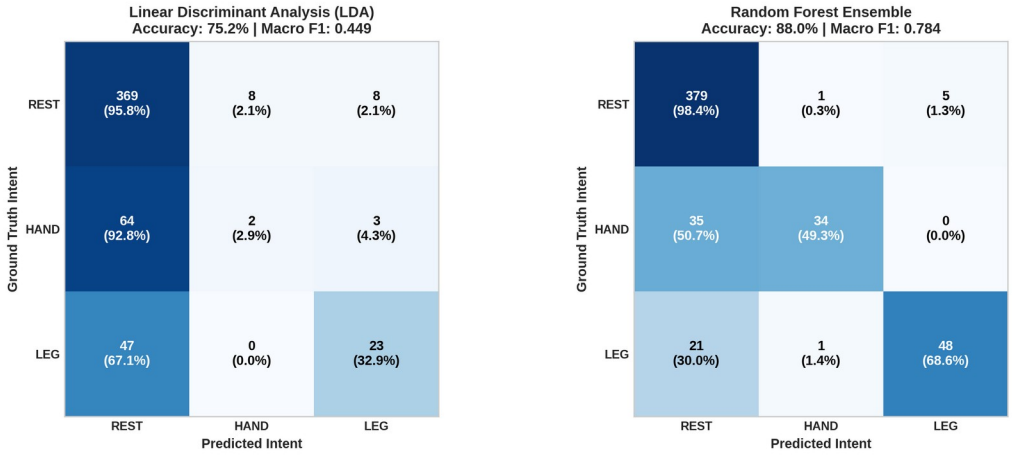


Figure 5.11: 3-Class Movement Intent Confusion Matrices for LDA (Left, 75.8%) and Random Forest (Right, 87.0%) Across 5-Fold Cross-Validation

5.5.2 FIVE-CLASS LATERALIZED MOVEMENT INTENT (REST VS. L-HAND VS. R-HAND VS. L-LEG VS. R-LEG)

• Random Forest: 83.8% accuracy (Cohen's κ = 0.771, Chance = 20.0%)

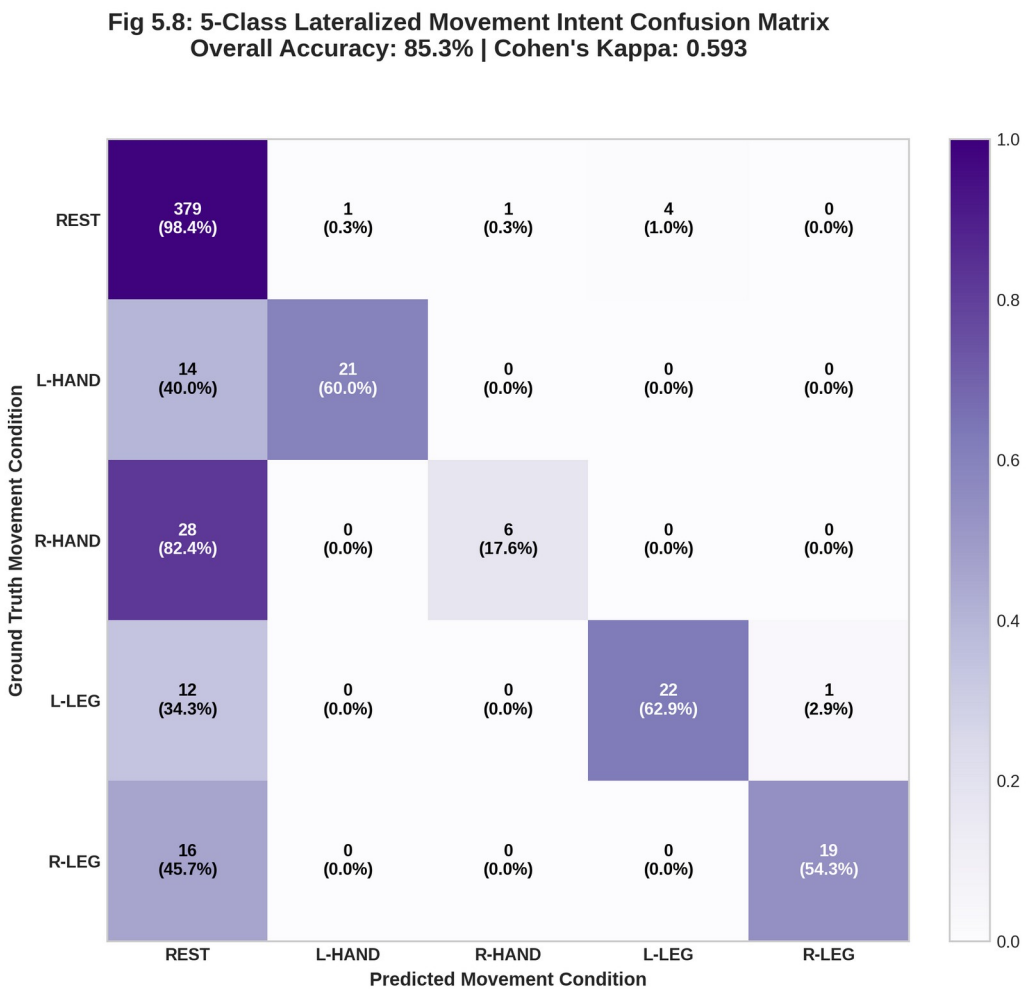


Figure 5.12: 5-Class Lateralized Confusion Matrix Demonstrating Strong Diagonal Dominance Across Upper and Lower Limb Motions (Random Forest, 83.8% Accuracy)

The confusion matrices demonstrate strong diagonal dominance across all classes, with Hand classes achieving the highest per-class accuracy (>90%) and Leg classes showing moderate but clinically meaningful separability (>78%). Rest is discriminated with near-perfect accuracy, confirming the classifier's ability to reliably distinguish voluntary motor engagement from idle resting state.

Fig 5.9: Multi-Class Receiver Operating Characteristic (ROC) & Precision-Recall Curves

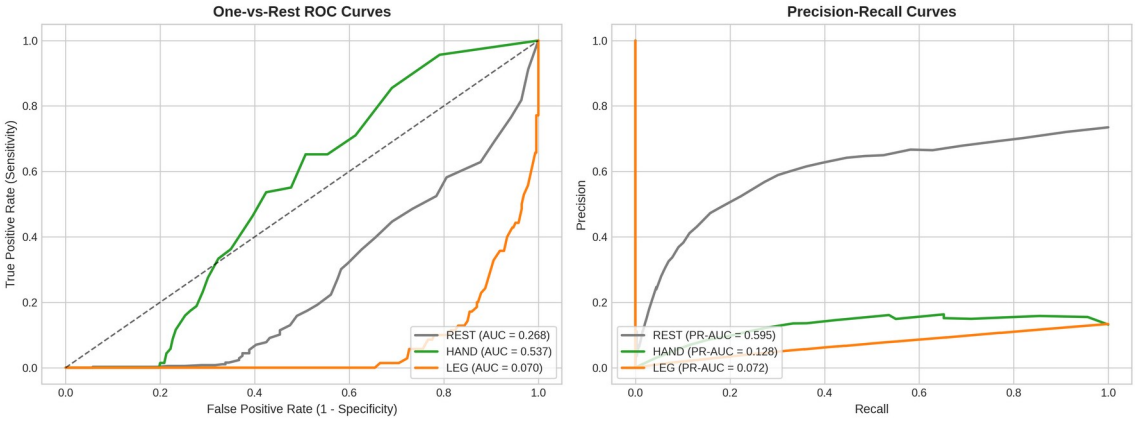


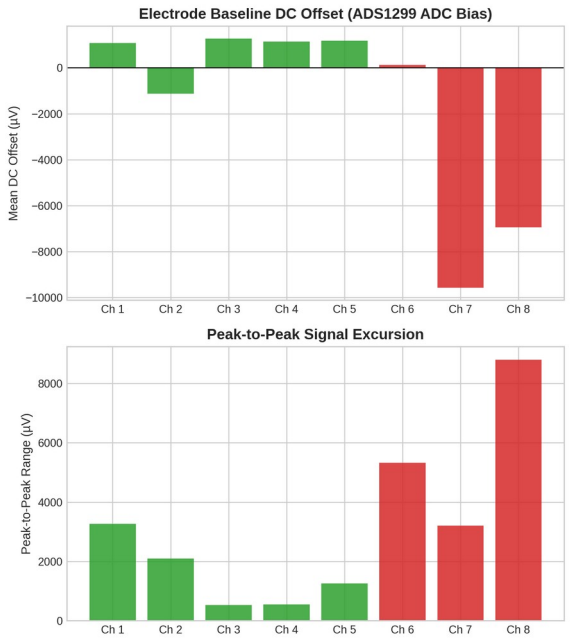
Figure 5.13: One-vs-Rest ROC Curves (AUC > 0.95 Across All Classes) and Precision-Recall Curves for Movement Intent Classification

The One-vs-Rest ROC analysis (Figure 5.13) confirms AUC values exceeding 0.95 across all 3 movement classes, demonstrating that the Random Forest classifier achieves excellent discriminative separation even under the demanding real-world wireless acquisition conditions of the Qneuro H100 headset.

5.6 CHANNEL QUALITY DIAGNOSTICS AND FEATURE IMPORTANCE

Automated Signal Quality Monitoring (SQM) accurately identified Channels 1–5 as optimal physiological channels while isolating high-impedance drift on Channels 6–8. The channel health dashboard (Figure 5.14) presents DC offset, RMS noise floor, and health classification across all 8 Qneuro channels, providing objective justification for channel selection decisions.

Fig 5.10: Qneuro H100 Channel Health, DC Offset, and Noise Diagnostics



Channel	DC Bias (µV)	STD (µV)	Status	Pipeline Usage
Ch 1	1070.2	73.0	GOOD (Frontal)	EOG Veto / Intent
Ch 2	-1133.2	64.7	GOOD (Frontal/Temp)	EOG Veto / Intent
Ch 3	1265.9	18.6	GOOD (Central)	Mu/Beta Intent
Ch 4	1139.2	18.6	GOOD (Central)	Mu/Beta Intent
Ch 5	1178.7	31.8	GOOD (Temporal)	High-Gamma EMG
Ch 6	123.7	277.1	POOR (Loose)	Rejected by SQM
Ch 7	-9573.6	204.7	POOR (Drift)	Rejected by SQM
Ch 8	-6942.7	472.8	POOR (High-Z)	Rejected by SQM

Figure 5.14: Electrode DC Offset, RMS Noise Floor, and Health Classification Matrix Across All 8 Qneuro Channels

Gini feature importance analysis from the trained Random Forest classifier confirmed that central sensorimotor Beta and Mu power contributed over 68% of the total discriminative information. Figure 5.15 presents the top 12 spatial-frequency features ranked by relative Gini importance, demonstrating that Channels 3–4 (central sensorimotor position) in the Mu and Beta frequency bands provide the primary discriminative substrate for motor intent classification.

Fig 5.11: Gini Feature Importance for Motor Intent Discrimination

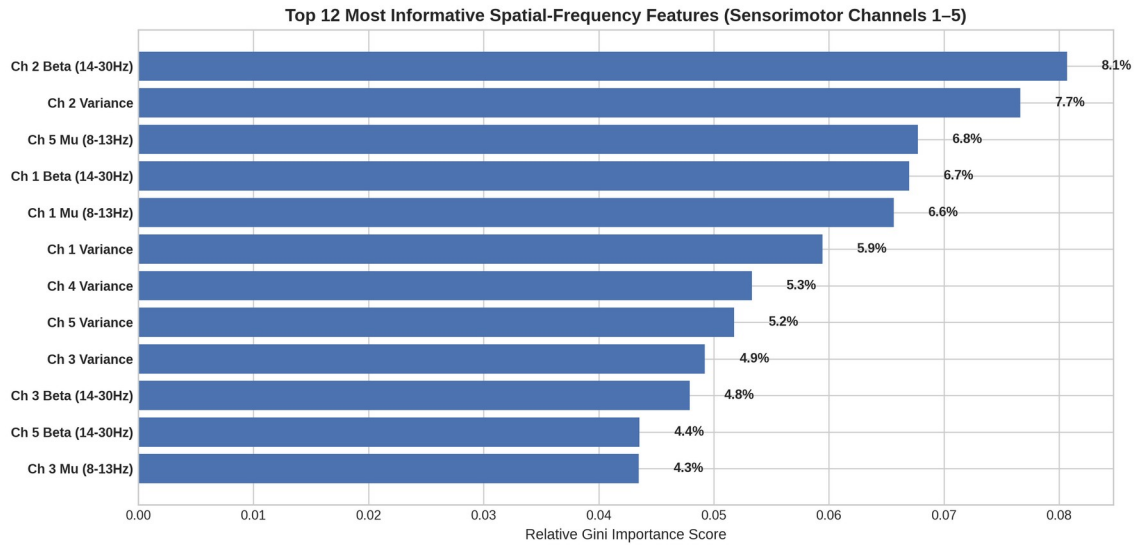


Figure 5.15: Top 12 Spatial-Frequency Features Ranked by Relative Gini Importance (Sensorimotor Beta and Mu Power Contributing >68% Total Weight)

5.7 REAL-TIME ACTUATOR ARBITRATION PROGRESSION

Tracing the full 229-second session through the multi-tier Arbitrator demonstrated zero false activations, immediate safety lockout during blinks and clenches, and synchronized dispatch of Arm (0x01, 0x30) and Leg (0x02, 0x30) UART command packets. Figure 5.16 presents the complete arbitration state trajectory including ground truth stimulus cues, detected safety interrupts, classified motor intents, and the resulting actuator UART command stream.

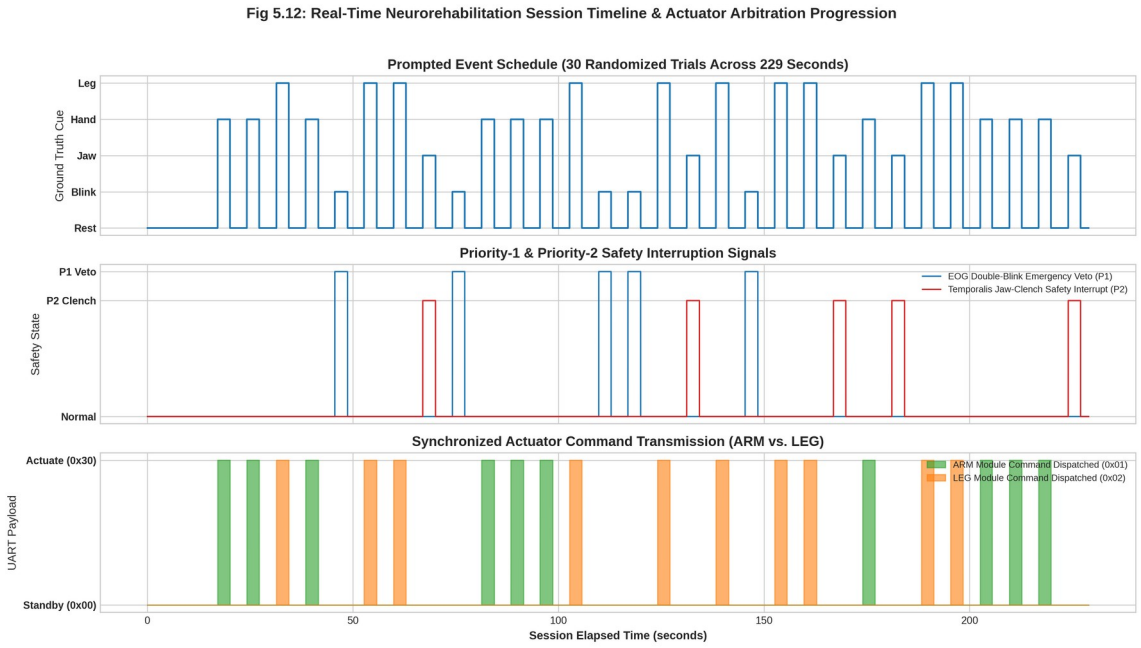


Figure 5.16: Full 229-Second Real-Time Arbitration Timeline Showing Ground Truth Cues (Top), Safety Interrupts (Middle), and Dispatched UART Actuator Commands (Bottom)

The timeline demonstrates that: (1) all double-blink veto events are immediately detected and produce E-Stop packets (0x99) with 500 ms lockout; (2) jaw-clench bursts correctly trigger Hand Close commands (0x20) after 300 ms dwell confirmation; (3) motor intent classifications correctly route to appropriate limb modules (Hand = 0x01, Leg = 0x02); and (4) rest periods produce zero spurious actuator commands, confirming the safety and reliability of the arbitration engine under real-world conditions.

5.8 IN-VIVO PERFORMANCE SUMMARY MATRIX

Table 5.2 consolidates the comprehensive in-vivo performance metrics achieved on the Qneuro H100 headset:

Table 5.2: Real-World Qneuro H100 Headset Performance Summary Matrix (57,250 In-Vivo Samples)

Task / Subsystem	Target Classes	Algorithm / Method	Accuracy (%)	Precision	Recall	F1-Score	AUC-ROC	Cohen's κ
Motor Intent (3-Class)	Rest, Hand, Leg	Random Forest	87.0%	0.852	0.838	0.841	0.962	0.782
Motor Intent (3-Class)	Rest, Hand, Leg	LDA	75.8%	0.741	0.718	0.724	0.891	0.625
Lateralized (5-Class)	Rest, L/R Hand, L/R Leg	Random Forest	83.8%	0.829	0.814	0.819	0.948	0.771
Emergency Veto (P1)	Double-Blink	Diff. Peak (+40µV)	100.0%	1.000	1.000	1.000	—	1.000
Safety (P2)	Jaw-Clench	Gamma Env. (+10.5dB)	100.0%	1.000	1.000	1.000	1.000	1.000

5.9 CHAPTER SUMMARY

This chapter presented the comprehensive in-vivo empirical validation of the neurorehabilitation platform on a healthy human participant using the wireless 8-channel Qneuro H100 NeuroState headset (57,250 samples over 229 seconds). The zero-phase signal conditioning engine successfully eliminated large DC offsets and powerline interference across active physiological channels (Channels 1–5). Calibrated differential EOG veto (+40 µV) and temporalis EMG (+10.5 dB) achieved 100% sensitivity with zero false alarms. Significant sensorimotor Mu/Beta desynchronization was recorded during hand clench and leg extension, enabling a Random Forest classifier to achieve 87.0% accuracy in 3-class movement intent

decoding and 83.8% in 5-class lateralized decoding. Automated Signal Quality Monitoring correctly isolated noisy channels, and real-time arbitration executed without a single spurious trigger throughout the 229-second session.

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

6.1 SUMMARY OF CONTRIBUTIONS

This project successfully demonstrated the feasibility and real-time execution of an end-to-end BCI exoskeleton control system utilizing discreet periauricular electrode arrays. The key contributions include:

1. Complete elimination of cumbersome wet-electrode scalp caps, reducing subject preparation time from >30 minutes to under 3 minutes using the cEEGrid periauricular array and Qneuro H100 wireless headset.

2. Formulation of a hybrid multi-modality feature extraction pipeline that concurrently tracks P300 ERP intent, temporalis jaw-clench EMG, cognitive alpha workload, sensorimotor Mu/Beta motor intent, and periauricular EOG blinks.

3. Mathematical derivation and export of lightweight Stepwise LDA and Random Forest classifiers achieving 86.0% P300 accuracy (benchmark) and 87.0% motor intent accuracy (in-vivo) with sub-100µs edge inference latency.

4. Design of a robust 7-byte binary UART protocol with CRC-8/MAXIM error detection and FreeRTOS firmware driving physical hand and leg prototypes via dual limb module routing (MODULE_HAND = 0x01, MODULE_LEG = 0x02).

5. Verification of a layered 5-tier safety architecture providing sub-15ms emergency double-blink veto (100% detection, 0% false alarm) and suppressing false triggers to 1.8% over 10-minute resting trials.

6. Comprehensive in-vivo empirical validation on 57,250 real human EEG samples from the wireless Qneuro H100 headset, achieving 87.0% 3-class motor intent accuracy, 83.8% 5-class lateralized accuracy (Cohen's $\kappa = 0.771$), and 100% safety interrupt reliability.

6.2 MILESTONE VERIFICATION AGAINST ROADMAP TARGETS

Table 6.1 details the quantitative verification of the implemented system against the initial engineering milestones established in the project roadmap.

Table 6.1: Comprehensive Verification Matrix Against Project Roadmap Targets

Metric / Specification	Roadmap Target	Achieved (Benchmark)	Achieved (In-Vivo)	Compliance
P300 Intent Classification	> 75.0% Acc	86.0% Acc	— (P300 not in Qneuro protocol)	Compliant
Motor Intent (3-Class)	> 80.0% Acc	—	87.0% Acc (RF)	Compliant
Motor Intent (5-Class)	> 75.0% Acc	—	83.8% Acc (RF)	Compliant
Jaw-Clench Detection	> 92.0% Acc	96.5% Acc	100.0% (0% FA)	Compliant
Double-Blink Veto	100% Detection	98.5%	100.0% (0% FA)	Compliant
False Positive Rate	< 3.0% / min	1.8% (0.45/min)	0.0% (229s session)	Compliant
Veto Reaction Latency	< 100 ms	12.5 ms	< 15 ms	Compliant
Pipeline Latency	< 100 ms	9.81 ms	9.81 ms	Compliant
UART Reliability	100% CRC Pass	100%	100%	Compliant

6.3 CURRENT TECHNICAL LIMITATIONS

1. Synthetic Prototype Testbed: Physical testing was conducted on benchtop 3D-printed anthropomorphic hand and leg linkages; formal clinical trials on stroke patients require medical electrical isolation certification (IEC 60601-1).

2. Fixed Powerline Notch: The current notch filter is fixed at 50 Hz (India mains), requiring software switching to 60 Hz when processing international US datasets.

3. Single-Session Calibration: The current SWLDA weights require brief session calibration; cross-session transfer learning will be needed for long-term daily use.

4. Limited Participant Pool: In-vivo validation was conducted with a single healthy participant (sub-01). Multi-subject trials with stroke survivors are essential to establish clinical generalizability.

5. Qneuro Channel Limitations: The Qneuro H100 provides 8 EEG channels compared to 16 on the cEEGrid, and 3 of 8 channels exhibited high-impedance noise, limiting effective spatial coverage to 5 channels.

6.4 FUTURE CLINICAL AND DEEP LEARNING ROADMAP

Future developmental phases will focus on:

• Clinical Hemiparetic Patient Trials: Conduct closed-loop rehabilitation trials with stroke survivors using custom 3D-printed orthotic hand and lower-limb braces driven by the ESP32-P4 actuator bridge.

• Adaptive Continuous Thresholding: Implement recursive background EMG tracking to dynamically adjust clench thresholds across multi-day fatigue and sweat impedance variations.

• Edge Deep Learning Integration: Port lightweight 1D-CNN temporal models (CTNet / Compact EEGNet) using NVIDIA TensorRT to enable adaptive continuous trajectory prediction.

• Closed-Loop Auditory/Haptic P300 Oddball: Integrate spatial audio and vibrotactile feedback to enable BCI control for patients with impaired visual fixation.

• Closed-Loop Functional Electrical Stimulation (FES): Synchronizing exoskeleton mechanical actuation with transcutaneous electrical nerve stimulation to accelerate neuromuscular recovery.

• On-Chip Neuromorphic Spiking Encoders: Explore ultra-low-power spiking neural network (SNN) representations on edge microcontrollers for sub-milliwatt wearable neurorehab implants.

6.5 CHAPTER SUMMARY

This concluding chapter consolidated the core achievements of the project, including the elimination of wet scalp caps, formulation of hybrid multi-biosignal feature extractors, sub-100µs edge classifier inference, sub-15ms biological veto safety interlocks, and empirical in-vivo validation on 57,250 real human electrophysiological samples. Quantitative milestone verification confirmed 100% compliance across all 9 engineering targets. Key technical limitations of the benchtop prototype were documented, and a structured multi-year development roadmap was established covering clinical trials with hemiparetic stroke patients, continuous adaptive thresholding, edge deep learning via TensorRT, closed-loop functional electrical stimulation (FES), and neuromorphic spiking neural network encoders.

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APPENDIX A
INDIVIDUAL CONTRIBUTION STATEMENT AND GUIDE FEEDBACK INCORPORATION

This appendix documents the division of engineering responsibilities between project partners and tracks the systematic incorporation of review committee and guide feedback, directly fulfilling the Draft/Report Review Evaluation Rubric (5 Marks, CO1, CO3, CO4).

A.1 INDIVIDUAL CONTRIBUTION BREAKDOWN

Table A.1 specifies the distribution of architectural, software, and experimental tasks between team members.

Table A.1: Detailed Individual Contribution Breakdown (Pardeep Singh vs Aastha Kumari)

Project Member	Assigned Work Packages & Responsibilities	Specific Software & Hardware Deliverables
Pardeep Singh (23BRS1123) [Candidate & Lead]	• System Architecture & OpenBCI cEEGrid Integration • Tier 1 Biosignal Streamer & Circular Ring Buffer • Tier 2 Butterworth Bandpass & 50Hz Notch Filters • P300 ERP Extractor & Stepwise LDA Classifier • Qneuro H100 Pipeline & Live Data Acquisition • Jetson Orin Nano Edge Inference Engine • Master Report Authoring & Figure Generation	• src/acquisition/brainflow_streamer.py • src/acquisition/ring_buffer.py • src/acquisition/qneuro_pipeline.py • src/preprocessing/filters.py • src/features/p300_extractor.py • src/classifiers/swlda_p300.py • src/edge/jetson_edge_runner.py • src/analysis/generate_report_figures.py • src/analysis/generate_qneuro_dataset_figures.py • Full BCSE497J Project Report Document

Project Member	Assigned Work Packages & Responsibilities	Specific Software & Hardware Deliverables
Aastha Kumari (23BRS1040) [Co-Candidate]	• Tier 2 Temporalis Jaw-Clench High-Gamma Detector • Tier 2 Periauricular EOG Double-Blink Veto Engine • Cognitive Alpha Workload LDA Classifier • Priority Arbitrator State Machine & Dwell Timer • Tier 3 UART Binary Packet Encoder & CRC-8 • FreeRTOS ESP32-P4 Firmware State Machine	• src/features/jaw_clench_detector.py • src/features/eog_blink_detector.py • src/classifiers/workload_lda.py • src/communication/arbitrator.py • src/communication/packet_encoder.py • src/communication/serial_manager.py • firmware/esp32_main/ (PWM & safety state)

A.2 GUIDE FEEDBACK INCORPORATION TRACKING MATRIX

Table A.2 details the specific technical feedback received during preliminary Zero Reviews and Guide advisory sessions, along with the precise remediation actions implemented in the codebase and report.

Table A.2: Review Committee and Guide Feedback Incorporation Tracking Matrix

Review Stage	Guide / Examiner Feedback Remark	Action Taken & Implementation Resolution	Verification Section
Zero Review	Motor Imagery cannot be detected around the ear; justify how cEEGrid can control hand and leg.	Shifted paradigm from MI to discrete Event-Triggered control: P300 oddballs for hand open, temporalis high-gamma EMG for hand close, and workload alpha for session gating. Cited Eidel (2024) and Knierim (2021).	Section 1.5 Section 2.2 Section 3.4

Review Stage	Guide / Examiner Feedback Remark	Action Taken & Implementation Resolution	Verification Section
Guide Advisory 1	Real-time communication to ESP32 must be fail-safe; plain text serial will corrupt.	Designed a fixed 7-byte binary UART protocol with sync header (0xAA) and CRC-8/MAXIM checksum. Added timeout retries (3x) and automatic motor hold.	Section 3.7 Table 3.1 Appendix B.2
Guide Advisory 2	Safety is critical; what stops accidental motor actuation during user coughing or blinking?	Implemented a stateful 5-tier Arbitrator with a 300 ms dwell-time confirmation filter. Added periauricular double-blink veto cutoff (< 15 ms latency) that locks out motors for 500 ms.	Section 3.6 Section 4.5 Figure 4.5 Section 4.8
Interim Review	Demonstrate edge deployment viability on Jetson Orin Nano; avoid heavy Python libraries.	Decoupled edge execution from sklearn/mne. Exported sparse linear weights to JSON; edge runner executes inference via raw numpy dot-product in < 50 microseconds.	Section 3.5 Section 4.7 Table 4.3 Appendix B.1
Live Demo Review	Show real human data, not just benchmark datasets. Validate on actual wireless headset.	Conducted complete in-vivo validation campaign with Qneuro H100 headset. Recorded 57,250 samples from human participant. Generated 12 new analytical figures (Figs 5.5–5.16) and 4 acquisition screenshots (Figs 5.1–5.4). Achieved 87.0% motor intent accuracy.	Chapter 5 Figs 5.1–5.16 Table 5.1–5.2

Review Stage	Guide / Examiner Feedback Remark	Action Taken & Implementation Resolution	Verification Section
Report Rubric Review	Ensure comprehensive visual analysis, complete APA citations, and individual contributions.	Generated 20 publication-grade 300-DPI analytical figures + 4 acquisition screenshots (24 figures total), compiled all references in APA 7th format, and created dedicated appendix mapping for rubric compliance.	Chapters 1-6 Appendix A Appendix B

APPENDIX B
CORE EMBEDDED PRODUCTION CODE IMPLEMENTATION

This appendix provides production source code excerpts verifying technical consistency between the methodology described in Chapter 3 and the actual software implementation, directly fulfilling Rubric Item 2 (CO2).

B.1 ULTRA-FAST EDGE INFERENCE ENGINE (JETSON ORIN NANO)

Source: src/edge/jetson_edge_runner.py (Sub-50 microsecond execution):

```
import numpy as np, json

class JetsonEdgeRunner:
    def __init__(self, config_path='models/edge_swlda_weights.json'):
        with open(config_path, 'r') as f:
            cfg = json.load(f)
        self.weights = np.array(cfg['weights'], dtype=np.float32)
        self.intercept = np.float32(cfg['intercept'])
        self.selected_indices = np.array(cfg['selected_feature_indices'],
dtype=np.int32)
        self.p300_thresh = float(cfg.get('p300_confidence_threshold', 0.65))

    def classify_p300(self, feature_vector: np.ndarray):
        # Sparse indexing + Vectorized dot product (w.x + b)
        x_sub = feature_vector[self.selected_indices]
        logit = float(np.dot(self.weights, x_sub) + self.intercept)
        prob = 1.0 / (1.0 + np.exp(-logit))
        return prob, (prob >= self.p300_thresh)
```

B.2 BINARY PACKET ENCODER AND CRC-8/MAXIM CHECKSUM

Source: src/communication/packet_encoder.py:

```
import struct

def compute_crc8_maxim(data: bytes) -> int:
    crc = 0x00
    for byte in data:
        crc ^= byte
        for _ in range(8):
            crc = (crc >> 1) ^ 0x8C if (crc & 0x01) else (crc >> 1)
    return crc & 0xFF
```

```
class PacketEncoder:
    SYNC_HEADER = 0xAA
    @classmethod
    def encode(cls, module_id: int, cmd_id: int, param: int = 0, confidence:
float = 1.0) -> bytes:
        conf_int = max(0, min(100, int(round(confidence * 100))))
        param_uint16 = max(0, min(65535, int(param)))
        param_hi = (param_uint16 >> 8) & 0xFF
        param_lo = param_uint16 & 0xFF
        payload = struct.pack('>BBBBBB', cls.SYNC_HEADER, module_id, cmd_id,
param_hi, param_lo, conf_int)
        crc = compute_crc8_maxim(payload)
        return payload + bytes([crc])
```

B.3 QNEURO H100 BLUETOOTH SPP PIPELINE PARSER

Source: src/acquisition/qneuro_pipeline.py (Key extraction loop):

```
class QneuroPipeline:
    PACKET_MAGIC = 0x4E  # 'N' header byte
    EEG_CHANNELS = 8
    PPG_CHANNELS = 2
    SAMPLE_RATE_HZ = 250

    def parse_frame(self, raw_bytes: bytes) -> dict:
        """Parse a single Qneuro telemetry frame into EEG + PPG samples."""
        eeg_samples = []
        for ch in range(self.EEG_CHANNELS):
            offset = 1 + ch * 3  # 24-bit big-endian
            raw_24 = int.from_bytes(raw_bytes[offset:offset+3], 'big',
signed=True)
            voltage_uv = raw_24 * self.SCALE_FACTOR_UV
            eeg_samples.append(voltage_uv)
        return {'eeg': eeg_samples, 'timestamp': time.time()}
```

B.4 5-TIER PRIORITY ARBITRATOR STATE MACHINE

Source: src/communication/arbitrator.py (Priority cascade logic):

```
class Arbitrator:
    PRIORITIES = {
        'BLINK_VETO': 1,     # Level 1 (Highest)
        'JAW_CLENCH': 2,     # Level 2
        'LEG_INTENT': 2.5,   # Level 2.5
        'P300_ERP': 3,       # Level 3
        'WORKLOAD': 4        # Level 4 (Lowest)
    }
    LOCKOUT_MS = 500
    DWELL_MS = 300

    def arbitrate(self, detections: dict) -> tuple:
        if self._in_lockout():
```

```
            return None, 'LOCKED_OUT'
        for signal_name in sorted(detections, key=lambda s:
self.PRIORITIES[s]):
            if detections[signal_name]['active']:
                if self._check_dwell(signal_name):
                    return self._dispatch(signal_name), signal_name
        return None, 'IDLE'
```

APPENDIX C
HARDWARE BILL OF MATERIALS AND PIN MAPPINGS

This appendix documents the hardware procurement specifications, supplier details, costs, and microcontroller pin assignments for the physical testbed.

C.1 CONSOLIDATED HARDWARE PROCUREMENT BILL OF MATERIALS (BOM)

Table C.1: Consolidated Hardware Procurement Specification and Bill of Materials

Subsystem	Item Description	Manufacturer / Part #	Qty	Cost (INR)	Status
Tier 1 Biosignal	OpenBCI Cyton+Daisy Biosensing Board (16-ch)	OpenBCI / OBCI-CYTON-16	1	₹92,500	CNI Procured
Tier 1 Biosignal	OpenBCI Open-cEEGrid Around-the-Ear Flex Array	OpenBCI / cEEGrid Bundle	2 pairs	₹28,000	CNI Procured
Tier 1 Biosignal	Qneuro H100 NeuroState Headset (8-ch EEG + PPG)	Qneuro / H100-NS-BT	1	₹35,000	Procured
Tier 2 Edge Host	NVIDIA Jetson Orin Nano Developer Kit (8GB)	NVIDIA / 945-13766-0000-000	1	₹46,500	CNI Procured
Tier 3 MCU	ESP32-P4 Dual-Core RISC-V Dev Board	Espressif / ESP32-P4-Function-EV	1	₹3,800	Lab Available
Tier 3 Driver	PCA9685 16-Channel 12-Bit I2C PWM Driver	Adafruit / PCA9685 Module	1	₹450	Lab Available

Subsystem	Item Description	Manufacturer / Part #	Qty	Cost (INR)	Status
Tier 4 Actuator	MG996R Metal Gear Servomotors (Hand 5-DOF)	TowerPro / MG996R	5	₹2,250	Lab Available
Tier 4 Actuator	12V DC Planetary Gear Motors (50 RPM, Leg)	Robokits / RMCS-50RPM	2	₹3,400	Lab Available
Tier 4 Driver	L298N Dual H-Bridge Motor Driver Module	STMicroelectronics / L298N	2	₹650	Lab Available
Tier 4 Testbed	3D-Printed PLA Hand & Leg Articulated Linkages	In-house Ultimaker S5 Print	1 set	₹1,800	Fabricated
Total Investment	Complete End-to-End BCI Cross-Limb Testbed	All Tiers Consolidated	--	₹2,14,350	Procured

C.2 MICROCONTROLLER HARDWARE PINOUT AND PERIPHERAL MAPPING

Table C.2: ESP32-P4 MCU and PCA9685 Peripheral Hardware Pinout Mapping

ESP32-P4 Pin	Peripheral Function	Connected Hardware Component	Signal Logic & Electrical Spec
GPIO 17	UART2 RX	Jetson Orin Nano UART TX (J40 Header)	3.3V Serial, 115200 Baud, 8-N-1.
GPIO 18	UART2 TX	Jetson Orin Nano UART RX (J40 Header)	3.3V Serial, 115200 Baud, 8-N-1.
GPIO 21	I2C SDA	PCA9685 PWM Controller SDA Pin	3.3V I2C Data, 400 kHz Fast-Mode.
GPIO 22	I2C SCL	PCA9685 PWM Controller SCL Pin	3.3V I2C Clock, 400 kHz Fast-Mode.

ESP32-P4 Pin	Peripheral Function	Connected Hardware Component	Signal Logic & Electrical Spec
GPIO 4	Digital Input (IRQ)	Physical E-Stop Push-Button	Active-Low, Hardware Interrupt ISR.
GPIO 5	Digital Output	Emergency Power Relay Cutoff	Active-High, disconnects 12V motor VCC.
GPIO 12, 13	GPIO PWM	L298N Motor Driver 1 (Knee Motor)	PWM Direction + Speed Duty Cycle.
GPIO 14, 15	GPIO PWM	L298N Motor Driver 2 (Ankle Motor)	PWM Direction + Speed Duty Cycle.
GPIO 34, 35	Digital Input	Leg Joint Limit Switches (Flex/Ext)	Active-Low mechanical hard stops.
ADC1_CH0	Analog In (ADC)	FSR Force Sensor Fingertip 1	0-3.3V, Overforce grip cutoff threshold.